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Robotic Automation of Mechanical Verification

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Robotic Automation of Mechanical Verification



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Robotisk Automatisering av Mekanisk Verifiering

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Sammanfattning

Denna masteruppsats utforskar implementeringen av kollaborativa robotar i Elektas mekaniska produktverifieringsprocess. Elekta är en pionjär inom precisionslösningar för strålterapi. Det är av stor vikt att säkerställa att Elektas produkter överensstämmer med kriterierna för prestanda, säkerhet och kvalitet. Den befintliga verifieringsprocessen står inför utmaningar när det gäller resurshantering, ergonomi, effektivitet och testnoggrannhet.

Denna avhandling fördjupar sig i tre huvudsakliga forskningsområden, studien börjar med teoretisk forskning som undersöker Industri 4.0, automation och människa-robotsamarbete, med fokus på ABB GoFa Cobot. Följande forskningsområde analyserar den befintliga marknaden, avseende utforskning av olika robotgripare. Slutligen fokuserar det tredje forskningsområdet på användaren. Inklusive intervjuer och observationer med Elekta som testar anställda och branschexperter från ABB och Atlas Copco.

Denna uppsats undersöker insikter från teoretisk forskning, marknads- och användarundersökning för att identifiera nyckelfaktorer som påverkar utvecklingen av koncept, prototyper och modeller. Genom detta skapades styrkor, svagheter och en kriterievärdesmatris för olika gripalternativ för Elektas tester. Dessutom definierades testparametrar för att göra testet lämpligt för automatiserad Cobot-verifiering. Till sist undersöks slutsatser från användarforskningen i flera delar, inklusive effektivitet, säkerhet, ergonomi, begränsningar, robotik och industri. Examensarbetet fortsatte med en kreativ process innehållande utveckling och iterationer av koncept genom CAD-modellering, 3D-utskrift och Cobot-programmering för ett specifikt verifieringstest.

Resultaten inkluderar riktlinjer för implementering av Cobot inom industrin, som täcker valet av lämpliga tester för automatisering, val av gripdon och utveckling av fingrar anpassade efter gripdonet. Rapporten fördjupar sig sedan i de hållbara aspekterna relaterade till Cobot-implementering, genom att studera social, ekonomisk och miljömässig hållbarhet. Vidare diskuterades avgörande faktorer som skulle kunna påverka projektet.

Sammanfattningsvis tar denna masteruppsats framgångsrikt upp Elektas utmaningar för mekanisk produktverifiering genom den strategiska implementeringen av Cobots. Det förväntas öka effektiviteten, minska ergonomiska risker, förbättra resurshanteringen och öka testprecisionen. Denna forskning ger en plan för Cobot-implementering hos Elekta och positionerar företaget i framkanten av modern industriell teknik, vilket betonar potentialen för ytterligare utveckling inom detta område.

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Abstract

This master thesis explores the implementation of Collaborative robots into Elekta's mechanical product verification process. Elekta is a pioneer in precision radiation therapy solutions. Ensuring the compliance of Elekta's products with performance, safety, and quality criteria is of great importance. The existing verification process faces challenges regarding resource management, ergonomics, efficiency, and test accuracy.

This thesis delves into three main research areas, the study begins with theoretical research investigating Industry 4.0, automation, and human-robot collaboration, focusing on the ABB GoFa Cobot. The following research area analyzes the existing market, regarding the exploration of different robotic grippers. Lastly, the third research area focuses on the user. Including interviews and observations with Elekta testing employees and industry experts from ABB and Atlas Copco.

This thesis examines insights from theoretical, market, and user research to identify key factors influencing the development of concepts, prototypes, and models. By this, strengths, weaknesses, and a criteria value matrix were created for various gripper options for Elekta's tests. Furthermore, test parameters were defined to make the test suitable for automated Cobot verification. Lastly, the conclusion from the user research was examined in several parts, including, efficiency, safety, ergonomics, limitations, robotics, and industry. The thesis continued with a creative process that contains the development and iterations of concepts by CAD modeling, 3D printing, and Cobot programming for a specific verification test.

The results of this thesis include Cobot implementation guidelines, covering the selection of suitable tests for automation, gripper selection, and finger development. The report then delves into the sustainable regards related to Cobot implementation, studying social, economic, and environmental sustainability. Furthermore, decisive factors that could influence the project were discussed.

In conclusion, this master's thesis successfully addresses Elekta's mechanical product verification challenges through the strategic implementation of Cobots. It is expected to increase efficiency, reduce ergonomic risks, improve resource management, and increase test precision. This research provides a blueprint for Cobot implementation at Elekta and positions the company at the forefront of modern industrial technology, emphasizing the potential for further development in this field.

Keywords

Cobot = Collaborative robot.

ASI = Arm side interface.

Wizard easy programming = Drag and drop block programming.

Lead through = Programming through human demonstration.

CAD = Computer Aided Design, for the design of digital 3D models.

CT Box = computed tomography box.

3D printing = Printing 3D models.

VR = Virtual reality

AR = Augmented reality

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1 Introduction

This introductory chapter provides an overview of Elekta's focus on developing precision radiation therapy solutions and their need for product verification to ensure compliance with performance, safety, and quality criteria. The current verification process faces challenges in terms of resource management and ergonomics. The goal was to investigate the potential areas where Cobots can be integrated into Elekta's product verification process, optimize tests, design a fixture, and demonstrate the advantages of employing Cobots. The thesis also considers the benefits, ethics, and sustainability aspects of the project. Overall, this chapter sets the foundation for the investigation into the incorporation of Cobots in Elekta's mechanical verification, outlining the objectives and structure of the thesis.

1.1 Background

Elekta is a leading innovator of precision radiation therapy solutions and works to ensure that every patient has access to the best cancer care possible. The company's primary focus is to develop and provide innovative treatment solutions for cancer and brain disorders. Elekta's product portfolio is designed to enhance the accuracy, effectiveness, and safety of cancer treatment, thereby improving patients' outcomes and quality of life. To verify that Elekta's products align with the necessary performance, safety, and quality criteria, product verification is of great importance.

Verification of products plays a crucial role in the evaluation and release process. It serves as a means to demonstrate compliance with requirements through various methods such as testing, analysis, inspection, or demonstration [33]. By conducting verification tests, Elekta can address requirements associated with risks and other aspects, ensuring that all products meet the necessary criteria. The requirements often pertain to the durability of the product, ensuring its optimal functioning throughout its lifespan. The testing procedure involves using the product as intended, which often involves repetitive movements and

evaluating its performance at different stages of its life, ensuring its durability and functionality. Elekta employs additional testers to execute these repetitive movements to facilitate this process, followed by verification engineers who conduct thorough product testing. Elekta has employed this approach for many years.

A collaborative robot, commonly known as a Cobot, possesses the capacity to operate close to humans in a secure manner. Its primary purpose is to facilitate joint efforts between individuals and the robot to carry out routine tasks. Cobots are frequently associated with enhanced productivity, as they possess remarkable precision when executing repetitive tasks. The utilization of Cobots in various industries holds significant potential for automation. One notable advantage of employing a Cobot for accuracy assessment lies in its ability to consistently conduct tests under controlled conditions with identical or varying force, speed, and movement [3].

1.2 Problem

Elekta is facing challenges in the mechanical product verification process, characterized by challenges in resource management, safety, and test efficiency. Certain tests require precise and controlled movements, which prove difficult to achieve with human testers, impacting overall test efficiency. Furthermore, managing and ensuring the availability of testers proves to be a hurdle, compounded by tests that often contain monotonous and repetitive movements that cause ergonomic problems. In response to these challenges, Elekta has made an investment in a Collaborative Robot, commonly referred to as a Cobot, with the aim to optimize its product verification process.

1.3 Purpose

This master thesis will serve as a foundation for Cobot implementation in mechanical verification processes, studying both today's product verification state

and the future potential of automation.

1.4 Goal

The objective of this thesis is to investigate the utilization of Cobots in mechanical product verification. It seeks to identify the automation potential of different mechanical verification tests, along with the following procedure of gripper selection and finger design for optimal Cobot implementation. The goal is to automate a complex mechanical verification test to gain valuable insight into the implementation process and highlight the following advantages of employing Cobots.

1.5 Deliverables

This master thesis will deliver:

- A comprehensive study and analysis on Cobot utilization and limitations with today's product verification process.
- A complete automation of a complex mechanical verification test, selected from Elekta's ongoing tests.
- Guidelines regarding future Cobot implementation.
- Verification of the implemented solution through testing precision and performance, along with an examination of its advantages, ethical considerations, and sustainability.

1.6 Methodology

The Master Thesis Project will be divided into four main parts: extensive research, analysis, design work, and report writing. The literature study will be performed first, to be able to better understand mechanical product verification with the use of a Cobot. The analysis will define the key insights gathered during the

extensive research. The design work will include the generation of ideas, the creation of mock-ups and concepts, testing, evaluation, validation of concepts, CAD, 3D printing, and Cobot programming generating an automated product verification process for a certain test. The report will be written continuously during the project, and the last few weeks will be used for final adjustments. For a more detailed method overview, view GANTT schedule in Appendix A.1.

1.7 Stakeholders

These stakeholders, with their expertise, collaboration, and support, are essential for mapping and utilizing Cobot for the automation of Elekta's mechanical product verification process, and essential to meet the goals of this thesis.

1.7.1 Elekta

A tight collaboration with Elekta will be essential for gaining valuable insights and optimizing results. Key stakeholders within Elekta are presented below.

- Quality Assurance and Testing Team
  This team is directly involved in the product verification process. Their expertise and experience are essential for a successful Cobot implementation and will influence how the implementation can improve the accuracy and efficiency of testing procedures and reduce human errors and risks.

- Employees and workers
  Testing personnel are key stakeholders whose feedback, concerns, and experience will shape the implementation of Cobots in the product verification process, ensuring a smooth transition and a successful integration in terms of improved product quality and safety.

1.7.2 External experts

The involvement of external experts is also essential. Their experience in Cobot implementations, automation, and the specific industry in which Elekta operates will aid with important insights to widen perspectives as well as knowledge.

1.8 Delimitation

The project has several limitations to consider. Firstly, the resources are limited to the utilization of ABB's GoFa Cobots, combined with the RG6 OnRobot parallel finger gripper. Elekta already purchased these resources, therefore they will be analyzed during the thesis. The thesis will assess the potential for automation in the Leksell Vantage system, selected in collaboration with Elekta while excluding consideration of alternative treatment solutions and verification processes. Time and resource constraints can limit the depth of research and implementation opportunities.

2 Methodology

Presented in this chapter are the different methods that have been used throughout the project to obtain the final results. The overall methodology structure follows an agile design process, structured as illustrated in Figure 2.1. An agile process model presents an iterative framework used for development processes. An iterative design process starts from the first stage and moves toward the last, with multiple interactions analyzing and further improving the output. The agile model was developed as a structure reference for the thesis work. The process involves the overall stages of research, analysis, creative process, and results, with the iterative loop integrated into the creative process. This section presents the methods used throughout the whole agile process. [20]

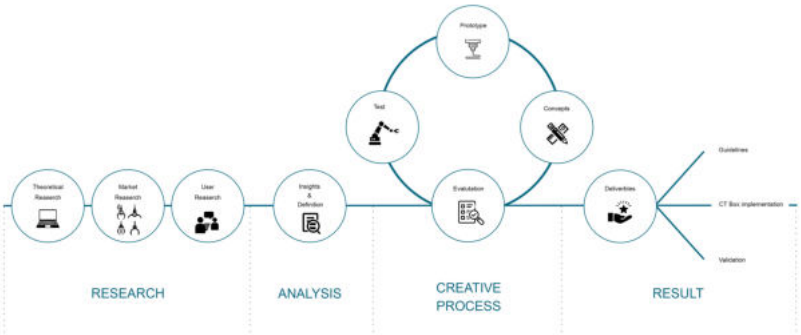


Figure 2.1: Agile process model

2.1 Research

The objective of the research, which is the initial phase of the mechanical design process, is to identify the need for a new product or area of development. The source behind automating mechanical verification is rooted in a technological push. This indicates that the concept of enhancing the product verification process is driven by new technology, rather than being pushed by customer or

user demand. [40] Technology push implies creating value through technological innovation, aiming to enhance processes to address future user needs. The advantages of technology push include accelerating technological progress and driving the company's economic growth forward by solving industry bottlenecks. [43] Furthermore, this chapter will include the different methods used to gather an understanding of the project scope.

2.1.1 Market research

Market analysis was used at the beginning of this project to identify possibilities when implementing a Cobot into the development and testing process at Elekta. A market analysis is used to map a company's potential, to determine what they have and what they can achieve according to [44]
. The market analysis researched Elekta's potential in automation by thoroughly investigating the gripper market, particularly focusing on Elekta's selected gripper. This process gathers valuable information, which is analyzed and applied to optimize the implementation and utilization of the gripper in Elekta's tests. Furthermore, it not only improves the implementation but also presents potential gripper solutions for forthcoming testing processes, ultimately enhancing Elekta's position in the automation landscape.

2.1.2 Interviews

The interviews have given insights into users' experiences, perspectives, behaviors, and requirements concerning specific products or situations in this thesis. In the early phases of this project, the interviews presented valuable perspectives on product usage patterns, user interactions, motivational factors, and pain points. [44]

In this project, semi-structured interviews have been conducted. These interviews encompass a predetermined set of thematic areas and associated questions intended to generate targeted information, this is illustrated in figure 2.2. The following interview method adopts a more flexible approach, giving participants

the freedom to explain their experiences surrounding the chosen subject. This approach facilitates the opportunity for additional questions and the discovery of unexplored insights that might remain concealed within a strictly structured interview framework. [44]



Figure 2.2: Interviews

2.1.3 Observations

Observation serves as a valuable method for uncovering unexplored areas of growth, specifically for identifying user requirements within a given context. Through the process of observation, participants are allowed to see, listen, and ask about a user's chosen scenario, this is illustrated in figure 2.3. In this project, both user observations and cognitive walk-throughs have been conducted. User observation involves participants closely observing, listening, and meticulously documenting the user's interactions with the product, allowing the user to test the product according to a typical scenario. Meanwhile, a cognitive walk-through entails users executing each step in product usage and communicating their actions, during this process, participants can also pose predetermined questions about these steps. Observation gives the participants a deeper understanding of essential aspects of product usage. [44]



Figure 2.3: Observation

2.2 Analysis

This introduces the methodologies employed to analyze the acquired knowledge in this thesis. The analysis is used to evaluate, define, and discover insights from information obtained through theoretical and market research, user research, and Elekta's tests.

2.2.1 Weighted decision matrix

A weighted decision matrix is a strategic tool that facilitates well-informed decision-making by systematically evaluating options against predetermined criteria, especially in complex scenarios with several criteria. It incorporates diverse perspectives, aligns options with product specifications, offers an overview of information, provides support in complex decisions, and can function as a discussion framework. This approach streamlines the decision process, enhances transparency, and ensures that decisions are both comprehensive and aligned with defined goals. The utilization of this method aimed to assess the compatibility between gripper solutions and Elekta's various product verification processes. To give an insight into the choice of gripper depending on different test criteria.

2.2.2 User analysis

The data collected through interviews, and observations will be subjected to thorough analysis. Quantitative data will be processed using statistical methods, while qualitative data will undergo thematic analysis. This process will help in identifying common patterns, trends, and insights related to emotional, cognitive, and physical aspects that can inform the design and development phases of the research.

Understanding the users and their emotional, cognitive, and physical needs is essential in designing systems that are not only effective and user-friendly but also emotionally engaging, cognitively supportive, and physically accessible. This user analysis will provide valuable insights that can guide the development process, considering these holistic user needs. Additionally, it can help in identifying potential challenges and barriers early in the design phase, ultimately contributing to the success of the project.

The ergonomics will be thoroughly analyzed in this thesis. This involves the examination of human interactions with products, environments, or tasks to ensure optimal efficiency, comfort, and safety. It aims to identify potential issues that might lead to discomfort or injury. Methods used for the ergonomic analysis in this project are Borg's CR-10 and pain mapping, this is illustrated in figure 2.4. Borg's CR-10 scale is a numerical rating system in ergonomics that quantifies perceived pain levels during tasks, also reviewed when observing the use of products. Additionally pain map was used, which visually represents areas of discomfort or strain on the human body, helping to pinpoint problem areas and develop solutions that enhance usability and well-being. [37]

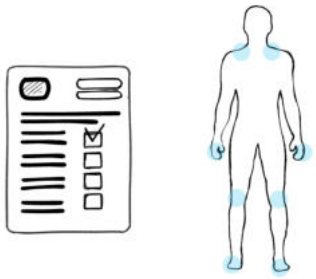


Figure 2.4: Ergonomic analysis

In the user analysis, the information obtained from the interviews was broken down, see illustration in figure 2.5. The process of analyzing user interviews is both giving and challenging. It concerns converting raw data into valuable and actionable insights, increasing certainty around the problem. [41] In this project, the interview analysis involves organizing post-it notes containing interview insights into thematic sections for sorting and categorization. Additionally, this enables the exploration of each section and its development of ideas.

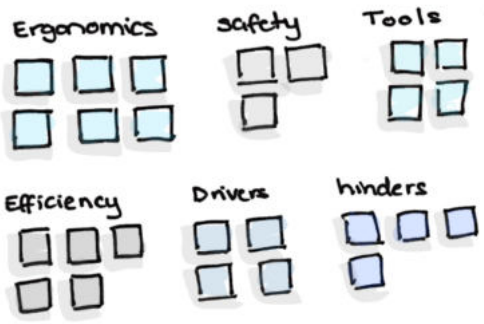


Figure 2.5: Interview analysis

2.2.3 Requirement specification

A requirement specification is a detailed document of criteria, functionalities, and limitations that a project or product has. Serving as a guideline, it directs the development process by setting expectations, preventing misunderstandings, and facilitating effective communication. This specification acts as a model for

evaluating the final product's alignment with goals and user needs, making it a crucial tool in ensuring successful and high-quality outcomes. In this project, this has been a crucial factor in the development and the final result of Cobot implementation.

2.3 Creative process

In the phase of the creative process, various methodologies were used to facilitate exploration and idea generation, aiming to address specific design challenges. Furthermore, this stage displays the methodologies used to create concepts that were developed into practical and effective solutions for the chosen Elekta test.

2.3.1 Brainstorming

Brainstorming is a method that generates a high volume of creative ideas for design challenges, this is illustrated in figure 2.10. It encourages unconventional thinking, emphasizes quantity over quality, and employs techniques like sketching and post-it notes to explore diverse solutions, according to [44]. The brainstorming method was an important catalyst for creativity and problem-solving in this project. Multiple brainstorming sessions were conducted to create a collection of creative solutions for carrying out the selected Elekta test. These sessions were instrumental in identifying unique approaches and refining the testing process. The integration of brainstorming not only fostered creativity but also significantly contributed to optimizing Elekta's Cobot.



Figure 2.6: Brainstorming

2.3.2 Crazy 8

Crazy 8s is a time-constrained brainstorming technique widely utilized in design projects and creative problem-solving, especially in fields like product- and user-experience design. This method prompts the participants to rapidly generate numerous out-of-the-box ideas within a certain time frame. Participants gather and each is given a sheet of paper that is divided into eight sections. A time limit of one minute is set for each sketch, and these sketches are targeted at a specific problem, striving to create new solutions. With each participant restricted to one minute, the focus is on quick, unconventional thinking. When the total time of eight minutes is up the sketches are shared and discussed, fostering open conversations and feedback. Then the sketches are passed to the next person and the procedure is done all over again but on four sketches of their choosing. This is to narrow down the ideas and to encourage fresh perspectives and potential concept refinement, the method is illustrated in different steps in figure 2.7. This method helps to ease the fear of judgment and gives freedom for creative ideas without overthinking. [12]. This method was used several times in this project to enhance creativity and to come up with a variety of solutions.



Figure 2.7: Crazy 8's

2.3.3 CAD

Computer-aided design (CAD) is software used for 3D modeling to create constructions and technical documentation, see illustration in figure 2.8. This is used to explore concepts and design ideas and visually be able to see the model in photo-realistic 3D renders, according to [13]. CAD was employed throughout the iterative process of this project to formulate 3D models of the concepts. Furthermore, it was used when generating 3D renderings, facilitating 3D printing, and ultimately shaping the final product for Elekta's application in real-world test scenarios.



Figure 2.8: CAD

2.3.4 Prototyping

Prototypes serve as tangible representations of ideas, facilitating the testing and realization of concepts, this is illustrated in figure 2.9. They function as educational tools, deepening one's comprehension of the underlying concept while uncovering opportunities for refinement leading to the final solution. In

the course of the project, prototyping has been used to evaluate, and test different concepts and to find valuable insights, leading to a final solution.



Figure 2.9: Prototyping

2.3.5 3D printing

3D printing is a free-form and additive manufacturing method that gives a lot of opportunities to create models and products. The technology produces 3D prototypes by vertically and horizontally printing layer after layer. The 3D prints are created from data files, often from CAD programs, and can consist of several different materials. [29]. This method was used in the creative process to create several Cobot fingers, this is illustrated in figure 2.10. This facilitated testing their compatibility with the Cobot and their functional performance in conjunction with Elekta's products.

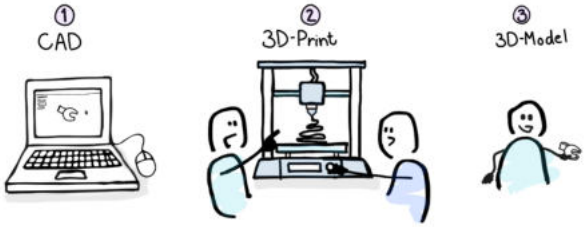


Figure 2.10: 3D-printing

2.3.6 Programming

Programming comprises a technological process that guides computers in executing specific tasks to address complex problems. This is seen as a

partnership between human creativity and computational prowess. In this dynamic, humans generate instructions, commonly called ”code”, which are comprehensible to computers. In the framework of the project, the programming was performed using the software RobotStudio® Suite, where the code was translated into executable commands for a Cobot. The RobotStudio® Suite allows users to employ various coding methods, including text-based programming, block-based programming, or a combination of both. [36] In contrast to conventional text coding, block coding languages entail pieces of code, each performing specific tasks, integrated into graphical blocks. Similar to Lego bricks, these blocks can be assembled in a specific order to compose a complete program, offering a more accessible way of programming. [5] The coding technique employed to program the Cobot in this project involved block programming through the FlexPedant, as illustrated in figure 2.11. This approach was complemented by text-based programming to refine the code and define more precise Cobot movements. These methods were used to develop the program for the final automated test of the Vantage Fiducial box and frame holder.

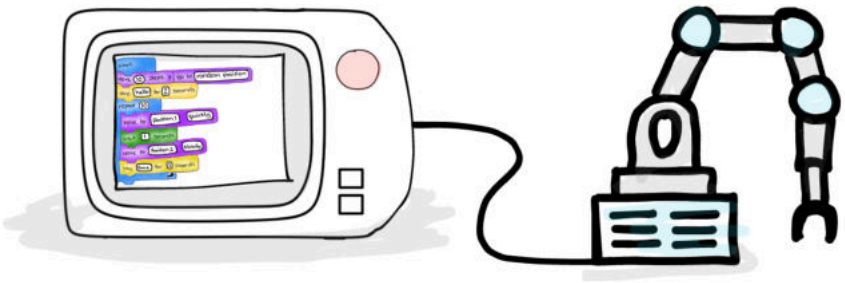


Figure 2.11: Programming

2.3.7 Cobot implementation guidelines

Developing implementation guidelines is a structured and essential process that provides a clear path for the execution of a project, a system, or a set of procedures. The aim is to ensure that all stakeholders involved in the implementation phase understand their roles, responsibilities, and the steps to follow.

3 Research

The research serves as a foundation for exploring the background and origins of the thesis. It provides a comprehensive exploration of Cobot-assisted product verification through three key research dimensions: theoretical, market, and user research. These research dimensions were studied to deliver an understanding of Cobot-assisted product verification and its potential for enhancing efficiency and productivity in diverse industries. Lastly to research potential areas of implementation at Elekta.

3.1 Theoretical Research

The theoretical research includes an in-depth literature study on Industry 4.0, automation, and human-robot collaboration, focusing on the key features and components of the ABB GoFa Cobot. Safety considerations in human-robot collaboration and mechanical verification at Elekta are also addressed. Key insights from the theoretical research phase are summarized and presented in chapter 4.1.

3.1.1 Industry 4.0

Industry 4.0, also known as the fourth revolution was introduced in 2011 by the German Federal Government [8]. The revolution introduces a new era of technologies including robotics, automation, artificial intelligence (AI), cyber-physical systems, the Internet of Things, etc. Robotics and Industry 4.0 aim to improve productivity while producing high-quality products [17]. Smart factories, which are the heart of Industry 4.0 provide an evolution in the value chain, where information and communication technology brings a much higher level of automation. The interconnection of humans, objects, and systems generates dynamic, real-time optimized, and inter-company value creation systems. An essential aspect of Industry 4.0 involves using autonomous methods driven by robots to intelligently accomplish tasks, with a primary focus on safety, flexibility, versatility, and collaboration. In Industry 4.0, robots and humans work in

collaboration, creating a safe human-machine interface. This interface can allow companies to set up 'lights out' factories where automated robots continue to work without light or heat after employee hours. Human resources can then be used more efficiently for other important tasks. The fourth industrial revolution characterized by cyber-physical systems, the Internet of Things, and the Internet of Services will assist companies to leverage productivity and economic gains [8]. These advantages align with Elekta's pursuit of industrial development. Therefore, a thorough understanding of Industry 4.0 is essential in the preliminary study for the robotic automation of mechanical verification.

3.1.2 The term automation

Automation involves the utilization of technology, including computers and robots, to carry out tasks that were previously performed by humans. Automation can be both physical and cognitive, with varying levels of advancement. Physical automation refers to the use of machines and robots to complete tasks, while cognitive automation involves providing information and support to human operators. Since the early 2010s, automation techniques have improved due to the technical revolution of Industry 4.0 and increased competition between companies. Companies, including Elekta, choose to automate to achieve better efficiency, quality, flexibility, time savings, and ergonomic benefits.[14]

3.1.3 Human-Robot Collaboration

Over the last decade, a new category of robots has been introduced to the market, collaborative robots, commonly known as Cobots. Cobots are designed to physically interact with humans in a shared environment, they combine the ability of a human to judge, react, and plan with the repeatability, precision, and strength of a robot.[28]. The primary principle underlying the Cobot is the possibility of processing a tremendous amount of information, understanding the surrounding environment, and responding to it while ensuring safety during direct human interaction. The fundamental distinction between traditional robots and Cobots lies in the Cobots' capability to adjust to new parameters and operate

in close proximity to humans without safeguarding the robotic system. [16]. Collaboration between humans and Cobots delivers a promising way to increase productivity while decreasing costs. [28]. The choice to implement a Human-Robot Collaboration is often motivated by economic or occupational health (ergonomic and human factors) aspects. The collaborative system harnesses a significant degree of flexibility, as the collaborative cells do not necessitate rigid safety systems. Consequently, they can easily and quickly be reallocated to alternative areas, enabling a great adaptation to systems that require continual changes. However, it should be noted that applications have to be constrained according to the international technical specification ISO/TS 15066:2016 'Robots and Robotic devices-Collaborative robots'[23], thus restricting flexibility. [28].

3.1.4 Mechanical verification using a Cobot

Introducing a Cobot into the product verification process at Elekta offers numerous advantages over traditional methods. The Cobot's tireless operation and avoidance of ergonomic risks increase efficiency, leading to higher productivity and faster testing cycles. The Cobot excels in executing the repetitive tasks of product verification while reducing human error and test result variability. Eliminating human error and variability Cobot implementation allows for improved test accuracy and quality control, with high-precision capabilities resulting in reliable and consistent data. By switching from traditional testing methods to Cobot verification with the ability to use lead-through programming, described in section 3.1.6, time resources can be optimized significantly. The Cobot flexibility allows for an effortless adaption to different verification processes. The Cobot can easily be programmed and adjusted to specific tests and setups. Utilizing Cobots for product verification and testing reduces labor costs by automating repetitive test procedures. [19] In relation to traditional robotic systems, Cobots are an affordable and time-efficient alternative. By integrating a Cobot into operations in contrast to traditional robotics, time resources are reduced from the typical set-up time of 18 months to approximately 1 hour. [18]

3.1.5 ABB collaborative robot

This part of the theoretical research study explores the concept of Cobots in depth. The study specifically focuses on ABB's GoFa Cobot, presented in 3.1, examining its key features, component description, user interface, additional attachments, and programming capabilities using the FlexPedant and Wizard applications. The study specifically aims to provide a comprehensive understanding of Cobots, with a focus on gaining deeper knowledge of ABB's GoFa Cobot. Elekta purchased this particular Cobot after extensive dialogue with the supplier, ABB. Throughout these conversations, Elekta detailed the intended utilization area of the Cobot within its testing environment. ABB's GoFa Cobot emerged as the recommended solution, aligning with the specified requirements.

A Cobot distinguishes itself from other robots by its ability to work closely with humans. This unique characteristic allows Cobots and humans to effectively accomplish shared tasks. Cobots are the ideal companion to increase company productivity since they can perform repetitive and ergonomically challenging tasks flawlessly and tirelessly unlike humans' limitations in terms of endurance and precision. By leveraging the capabilities of Cobots, organizations can allocate more time to critical tasks of higher significance instead.[3]

3.1.6 Key Features of ABB's GoFa Cobot

The ABB GoFa Cobot, presented in figure 3.1 with specifications defined in table 3.1, has a payload range from less than 0.5 kg to 5 kg, allowing it to handle smaller loads. With a maximum speed of 2.2 meters per second, the Cobot's 6-axis agility has the ability to move loads with a reach of 950 millimeters. The ABB GoFa Cobot is designed for ease of use, with features to communicate with the human collaborator. The Cobot is equipped with an interactive light communicating the Cobot status, integrated joint design with torque and positioning sensors, VR and AR integration and the possibility to be mounted at any angle.[3] The Cobot can be programmed with Wizard easy programming, which includes drag and drop-based programming software [4]. Lead-through programming is also an

alternative when working with the Cobot. It, as well as Wizard programming, allows an easy user interface. Lead-through operations, also referred to as hand guidance programming, involve programming industrial robots through demonstration. During programming, a robot operator will physically move the robotic manipulator through the waypoints of a desired task. [35] The Cobot is equipped with an arm-side interface (ASI) which is an ease-of-use feature for greater user interaction with the Cobot. With the help of ASI, the user can move the Cobot to different positions, save the positions, and then view and edit them in the wizard program.[3]

Table 3.1: Cobot Specification

ABB GoFa Cobot Specification	
Payload range	0.5 - 5 kg
Speed	2.2 m/s
Reachability	950 mm
Number of axes	6
Weight	27 kg
Protection	IP54
Mounting	Any angle
Controller	OmniCore C30
Customer power supply	24V/2A supply
Tool flange	Standard ISO 9409-1-50



Figure 3.1: ABB GoFa Cobot

3.1.7 Cobot components

The GoFa collaborative robot is designed to facilitate a close and safe collaboration between humans and robots. The system includes the GoFa CRB 15000 Cobot, which consists of a six-axis manipulator arm and a Cobot base. Additionally, the system incorporates a hand-held operator device known as the FlexPendant, along with an omnicore controller that enables motion control. The GoFa Cobot is connected to the omnicore controller via cabling.[3]

Moving further into the components of the GoFa Cobot, each of its six axes is equipped with an individual brake system. The GoFa Cobot does not include a gripping solution, although the end-of-arm tooling provides support for grippers with vision, vacuum, and servo functionality. The GoFa Cobot features an ISO 9409-1-50 flange [22], serving as the mounting plate that defines the primary dimensions, designation, and marking for a circular plate, acting as the mechanical interface. The Cobot base offers versatile mounting options, including table, wall, or ceiling installation. The Arm Side Interface (ASI) is composed of two configurable buttons and a status light indication, serving as a means to

communicate the Cobot's status.[3]

The hand-held FlexPedant operator is used to run programs, jog the manipulator, and modify the Cobot programs. The FLexPedant is equipped with an emergency stop, which immediately stops all movements of the Cobot system, a MultiTouch display supporting standard gestures, and a thumb button enabling lead-trough of the Cobot.[3]

3.1.8 Safety regarding human-robot collaboration

When using a collaborative robot in industrial environments, it is essential to prioritize safety to protect human workers, products, and the overall work environment. Safety requirements and precautions are presented below and will be taken into account while implementing human-robot collaborations.

ISO/TS 15066:2016 - *Robots and robotic devices, Collaborative robots*, specify safety requirements for collaborative industrial robot systems and their work environments. The specification was developed by the International Organization for Standardization (ISO) and focuses on establishing principles and requirements for the design and implementation of collaborative robot systems. [23]. The ISO 15066 standard outlines a set of guidelines essential for establishing a hazard-free collaborative work environment that ensures safe human-robot interactions. These guidelines should be taken into account when incorporating passive and active safeguards into the robot system or designing the collaborative workspace. [10]. Some key guidelines of the safety requirement include limitations regarding power and force limiting, impact sensing and detection, hand-guiding, and speed limitations, safety-rated monitored stop, risk assessment and validation, protective measures, and training.

ISO 15066 serves as a valuable reference for manufacturers, integrators, and users of collaborative robot systems, enabling them to design and implement safe and effective human-robot collaboration in various industrial and non-industrial

applications. It promotes the use of Cobots as a productive and safe tool in a shared workspace, facilitating increased efficiency and flexibility in modern manufacturing and automation environments.[10]

3.1.9 Ergonomic hazards in work environments

A comprehensive study into ergonomic hazards in the work environment was undertaken to identify the origin of aches and pains experienced during the verification of specific products at Elekta. The insights regarding ergonomic hazards at Elekta are detailed in section 3.3.6 where insights from the user study are supported by the theoretical research in the subsequent section.

Ergonomics involves the interaction between humans, technology, and organizations with the aim of optimizing health, well-being, and performance.[6] In a larger scope ergonomics examines human behavioral, psychological, and physiological capabilities and limitations. The basic premise of ergonomics includes that operation demand should not exceed the operator's capabilities and limitations, this is to ensure that the worker would not be exposed to stresses that can affect their safety and health, as well as the company's quality and productivity. Ergonomic implementations should remove barriers between quality, productivity, and human health by adapting systems and environments to individuals instead of the other way around. N. Jaffar et al. identify three categories of risk factors one of them, biomechanical exposure, includes the characteristics of mechanical verification. Biomechanical exposure includes repetitive motion, high forces, deviations from natural body alignments, and contact stress or pressure, all critical risk factors in workplace safety that increase the probability of ergonomic injuries to the musculoskeletal system. Musculoskeletal injuries include injuries to muscles, tendons, ligaments, and joints which arise when performing activities that strain the human body. [24]

Repetition is defined as performing the same motions with the same body parts excessively, with little chance for rest or recovery. Repeated similar motions performed over a period of time could cause over-extensions and overuse of

certain muscular groups, possibly leading to muscular fatigue. Generally, the greater the number of repetitions, the greater the degree of risk. However, the relationship between repetition and the degree of injury can be increased when other risk factors are also present in the activity. [24]

Force can be defined as the amount of physical effort required to perform a task. Exerting a force on an object, and performing tasks such as gripping, lifting, pushing, or pulling, may overload the human muscles, tendons, and joints. The amount of force depends on the grip type, a pinch grip produces 3-5 times more force on the tendons in the wrist than a grip with the whole hand. It also depends on the weight of the object, body posture, type of activity, and task duration. Injuries caused by force are often associated with shoulders, neck, lower back, forearm, wrist, and hands. Generally, the greater the force, the greater the degree of risk. Although, similar to repetition, the injury risk will be greater if other risk factors are also present in the activity. [24]

Posture encompasses the positioning of various body parts. It is essential to maintain proper alignment for optimal musculoskeletal health. An awkward posture emerges when body joints are bent or twisted beyond a comfortable range of motion, such as leaning sideways, bending down at a low level, or bending a wrist when rotating an object. Muscles, tendons, and ligaments are forced to work harder and may experience heightened stress. Every person has a neutral zone of movements for every joint in the body, performing a task with body segments outside this neutral zone increases the risk of injury. Static posture can also generate ergonomic discomfort when an individual's muscles remain in a fixed position for an extended period. Despite the position potentially being within the neutral zone, it can still result in aches and pains. [24]

Contact stresses are caused by grasping, balancing, or manipulating hard, sharp objects, equipment, or instruments. The muscles and tendons in forearms, wrists, or hands impinge when working against the sharp edges. The sharp edges create local contact stress that occurs when the object comes in contact with the skin.

Nerves and tissues beneath the skin can then be injured by the pressure, the contact stress is often more extensive when the contact area is protected with less protective tissue, such as the wrist, palm, or fingers. [24]

3.2 Market Research

The market research provides an extensive overview of end-of-arm tooling, specifically focusing on robotic grippers. The study examines various robotic grippers that are currently available on the market. Among the analyzed grippers is the OnRobot RG6 Finger gripper, which is used in conjunction with Elekta's Cobot. Key insights from the market research phase are summarized and presented in chapter 4.1.

3.2.1 Robot Gripper

In robotics, robot grippers, also known as end-of-arm tooling, are mechanisms designed to interact with the environment, enabling the Cobot to grasp and securely retain objects. When integrated with a collaborative robot arm, the gripper simulates the functionality of a human hand, combining the strength of an arm with the dexterity of a hand. This presents numerous operational possibilities.[25] A robotic finger gripper consists of multiple components that collaborate to manipulate objects. The specific design and components may vary depending on the manufacturer, model, and intended use. However, typical gripper elements, illustrated in figure 3.2 include the gripper bracket and fingertips. Furthermore, the structure and shape of the fingertips can be modified according to specific applications. Additionally, fingertip extensions may be added for the gripper to pick and place workpieces in tight spaces.[31] The selection of a gripper plays a central role in the application of robotics to various operations. The specific choice of grippers depends on several factors, first and foremost what kind of application will be handled, including the nature of the object, its size, shape, weight, and material, as well as the required level of precision.[39] Alongside this, the necessity of data feedback, as well as payload

capacity and operational space also needs to be considered [25]. It proves challenging to identify a robotic gripper that possesses sufficient flexibility to ensure efficient and reliable handling of all component types [39]. Therefore, this section provides an overview of various gripping solutions currently available in the market, along with the leading manufacturers associated with each solution.

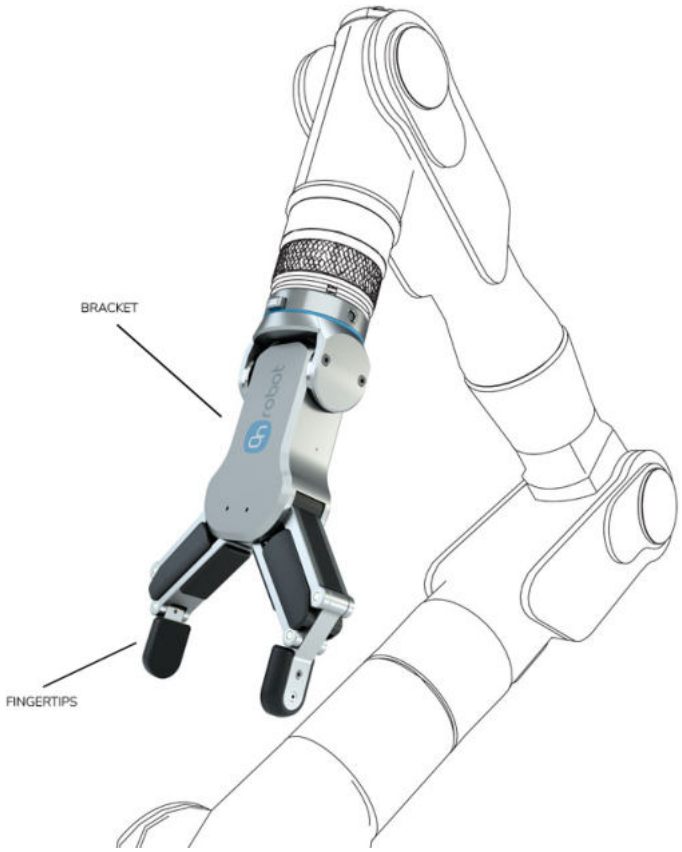


Figure 3.2: Onrobot finger gripper

3.2.2 Types of Grippers

In this section, a detailed description presents the state-of-the-art grippers available on today's market. Understanding the capabilities of a gripper is fundamental to optimizing its utilities in various tasks and facilitating an effective integration. Specifically, when choosing a gripper for the verification of medical products, careful consideration must be given to factors such as the product's dimensions, configuration, weight, material, and the precision level required. These considerations significantly influence the selection of an appropriate

gripper. To make well-informed decisions, it is important to conduct thorough research on the qualities of available grippers, a process outlined within this section. In section 4.1.2, a comparative analysis will be presented to assess and compare the capabilities of different grippers, providing straightforward means to evaluate and compare their respective features.

Finger Parallel Grippers

The standard mechanical grippers are finger and parallel-jaw grippers illustrated in Figure 3.3. They operate with two to four fingers, either electric or pneumatic-driven. The finger and parallel-jaw grippers are primarily used for smaller production tasks, they provide stability through applied force but lack adaptability to handle irregular shapes [11].

Two-finger and parallel-jaw grippers are ideal for regularly shaped parts, allowing customization for specific items. Three- and four-finger grippers excel with round or irregularly shaped objects [27]. However, parallel grippers may require custom jaws for specific items [1].

Finger grippers grasp items by operating with a pinching motion, requiring the items to be thinner than the gripper's maximum gap opening. This design limitation makes them rigid and less versatile. Two-finger grippers, in particular, struggle with heavy, long, or slippery objects due to insufficient friction for a secure grip. In such cases, a four-, six-, eight-, or more-fingered parallel gripper is necessary [1]. When designing a program for product verification, it's essential to factor in the capabilities of the finger and parallel-jaw grippers. Gripper customization may be needed if the product poses challenges to the gripper's standard handling abilities.

Major finger gripper manufacturers include OnRobot, Robotiq, WeissRobotics, Festo, and Piab [27].



Figure 3.3: Finger gripper

Vacuum Grippers

Vacuum grippers, illustrated in figure 3.4, use air pressure and suction cups to securely grip objects with an airtight seal, making them versatile for handling various materials and maintaining a high lifting capacity [27].

Compressed air-driven vacuum grippers excel at gripping and moving large, heavy objects, providing four to ten times more power than electromagnetic pumps. While commonly used for automating production, they are particularly suitable for packing and stacking tasks [1]. These grippers can also grasp a wide range of materials like plastic, glass, metal sheets, or wooden pallets [27].

However, their effectiveness relies on smooth, non-porous, flat surfaces, limiting their suitability for small components or irregularly shaped objects with limited planar surfaces to grip [21]. Despite being more affordable initially, their operational costs can be high due to substantial electricity consumption. Major vacuum gripper brands include OnRobot, Robotiq, Schmalz, Destaco, and Coval [27].



Figure 3.4: Vacuum gripper

Soft Grippers

Soft grippers, illustrated in figure 3.5, are designed for gently gripping delicate and breakable objects. Constructed from materials like silicon or molded plastic, these grippers offer a delicate yet secure touch, making them ideal for industries such as food, beverage, cosmetics, and pharmaceuticals.

These grippers are typically constructed using silicon or plastic molded material to ensure a delicate yet secure grip. Soft grippers excel at handling fragile objects in industries such as food, beverage, cosmetics, and pharmaceuticals. Their material minimizes the risk of breakage or damage, making them suitable for preventing surface scratches or marks [27].

Soft grippers can adapt to irregular shapes, and elastic or deformable materials, making them suitable for non-uniform or complex geometries and rubber components. However, their lower gripping strength makes them unsuitable for handling heavy objects, as this may compromise the grip's security [2]. Manufactured of soft grippers include OnRobot, Emi Corp, SoftRobotics, SoftGripping, and Ubiros [27].



Figure 3.5: Soft gripper

Magnetic Grippers

Magnetic grippers, illustrated in figure 3.6, are engineered to securely grip irregularly shaped metal parts using magnetic force, eliminating the need for air or power supply.[27] Primarily used for handling ferromagnetic materials, these grippers excel in manipulating thin, flat, and smooth materials without causing damage or deformation. Their simple design, with a single contact surface, ensures a rapid gripping process. Moreover, the magnetic gripper possesses the ability to grip parts with holes. Lastly, the low maintenance costs enhance the gripper's cost-effectiveness. [21]

Despite their advantages, magnetic grippers have some limitations. Objects may lose grip if moved too quickly, and even small amounts of oil can reduce the magnetic force on the surface. Machining debris may stick to the magnet during object removal, and the handled object may acquire residual magnetism. The grippers are only suitable for ferrous magnetic materials and have a weight limitation based on magnetic strength, making them unsuitable for handling heavy objects. Proper environmental control is also of great importance to avoid magnetic interference from nearby objects. [21] Major manufacturers of magnetic grippers include EMI, OnRobot, Schmalz, and SMC [27].



Figure 3.6: Magnetic gripper

Gecko Grippers

Gecko grippers, illustrated in figure 3.7, leverage innovative adhesive technology mimicking the natural mechanics of geckos. They utilize Van der Waals forces to attach to flat and smooth surfaces, mimicking the gecko's ability to adhere to diverse surfaces without external adhesives or suction.[30] This preload-hold-detach approach, where adhesive pads come into contact with the object, enables the gripper to pick up flat and porous objects without leaving marks. Gecko grippers are cost-efficient and reliable solutions for handling electronic components, metallic screws, or drilled materials. [27]

Gecko grippers excel at manipulating surfaces like glass, metal, plastic, and laminates, making them valuable in applications where a non-destructive and gentle gripping approach is essential [8]. However, their current limitation lies in the inflexibility of their backing, restricting their ability to conform to curved surfaces. Even with a softer backing, effective adhesion to curved edges remains a challenge [15]. Key manufacturers of Gecko grippers include OnRobot and Schunk [27].



Figure 3.7: Gecko gripper

3.2.3 OnRobot RG6 Finger gripper

After further dialogue with ABB, Elekta received a recommendation to acquire a parallel finger gripper to meet their requirements for handling diverse shapes with a strong grip, ensuring a firm and secure hold on manipulated objects. Elekta investigated the main manufacturers of finger grippers and chose the versatile OnRobot RG6 Finger gripper as the initial alternative for mechanical verification of their products.

The RG6 finger gripper from OnRobot, see figure 3.8 is a versatile and efficient tool designed for robotic automation. It is specifically designed to handle a wide range of objects with different shapes and sizes, making it suitable for various testing scenarios. [34]



Figure 3.8: RG6 finger gripper from OnRobot

The RG6 offers several advantages, firstly, it has a strong gripping force, allowing it to securely hold objects during tests, see table 3.2. It also features a wide opening range and adjustable fingers, which can be customized to match specific object geometries with varying sizes, ensuring a precise and reliable grip. Furthermore, the gripper is equipped with smart sensor technology to detect different objects and surfaces. This makes the gripper flexible in the way that it can sense how it should adjust to the test object, ensuring optimal performance and minimizing potential damage. [34]

Additionally, the RG6 features standardized mechanical and electrical interfaces, enabling it to be easily mounted and controlled by the Cobot. This seamless integration facilitates efficient and synchronized operation and makes for a plug-and-produce design which reduces employment time from one day to one hour. Lastly, the RG6 versatility and adaptability make it a valuable tool for automating various repetitive tasks in different industries. [34]. The specifications of the RG6 finger gripper are detailed in Table 3.2.

Table 3.2: RG6 Specification

OnRobot RG6 Specification	
Payload force fit	6 kg
Payload form fit	10 kg
Gripping speed	51 - 160 mm/s
Gripping force	25 - 120 N
Gripping time	0.05 - 0.15 s
Stroke length	160 mm
Dimensions	262 x 212 x 42 mm
Weight	1.25 kg
Protection	IP54
Finger position resolution	0.1 mm
Repetition accuracy	0.1 - 0.2
Power supply	20 - 25 V
Operating temperature	5-50 °C

The RG6 gripper offers customizable fingertips to accommodate various objects. The standard fingertips, see figure 3.9 made of rubber (EPDM), enhance the gripper's ability to handle different objects while minimizing marks. For cylindrical objects, the X-shaped fingertips, which are displayed in figure 3.9 C provide improved stability and payload capacity through force fit and form fit gripping approaches and are available as separate accessories. In addition, fingertip extensions of 50 and 100 mm, see figure 3.9 B are available to enable the gripping of objects in narrow spaces, such as boxes and crates.

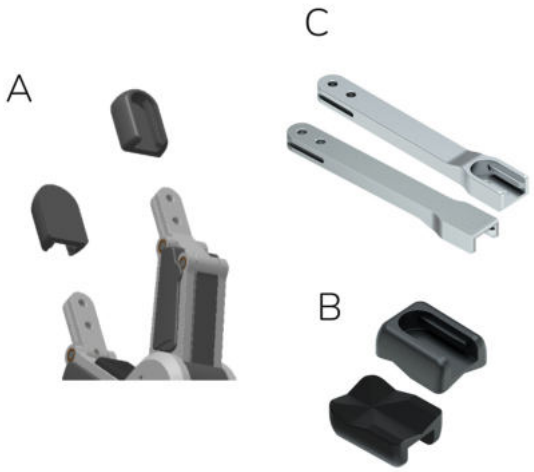


Figure 3.9: RG6 finger accessories

3.3 User research

The user research phase focused on gathering valuable insights regarding mechanical verification and the utilization of Cobots. Interviews were conducted to understand typical test procedures, tools, attachments, assess efficiency, safety, ergonomics, and uncover limitations in product development. Additionally, observations were made with Elekta employees and at Atlas Copco's manufacturing site. This comprehensive research aimed to contribute to the understanding of Cobot-assisted product verification and its potential. Shaping the resulting design and guidelines by incorporating industry expertise, ensuring effectiveness, and safety, and meeting Elekta's expectations. Key insights from the user research phase are summarized and presented in chapter 4.2.

3.3.1 Interviews

During the user research phase, interviews were conducted with Elekta testing employees and experts from companies working with Cobots. A total of eight interviews, six males and two females delivered a number of valuable insights presented in this section. See interview details in table 3.3. All interviews were conducted in a semi-structured manner. Initially, the interviews started off with predetermined questions outlined in appendix A.2. However, as the interviews progressed, participants were encouraged to engage in open discussion, and additional unplanned elaborative questions were posed. The following chapters present information gathered during the various interviews.

Table 3.3: Interview details

Interviewee	Gender	Age	Company	Role
1	Male	21 y/o	Elekta	Verification tester
2	Male	24 y/o	Elekta	Verification tester
3	Male	25 y/o	Elekta	Verification tester
4	Male	33 y/o	Elekta	Senior Verification Engineer
5	Female	58 y/o	Elekta	Lead Verification Engineer
6	Male	35 y/o	Atlas Copco	Business Manager
7	Female	51 y/o	Atlas Copco	Manager Assembly Engineering
8	Male	57 y/o	ABB	Team Lead Sales

3.3.2 Mechanical verification at Elekta

The process of mechanical verification at Elekta, illustrated in Figure 3.10, begins with a risk engineer defining the risks associated with the system under study. Subsequently, the verification engineers establish the system's requirements and develop a corresponding test case. After creating the test draft, it undergoes an evaluation against the verification readiness checklist (VR) to determine its readiness for execution. If the test draft successfully passes the VR evaluation, it proceeds to the testing phase. During testing, the testing space is first prepared and set up accordingly. Subsequently, documentation of testing procedures and measurements is undertaken and the findings are ultimately presented in a detailed test report. Once the testing procedure is completed, the test results are published in the traceability matrix (TMX). This matrix serves as a reference point

for all involved parties, enabling them to trace and review the outcomes of each test.

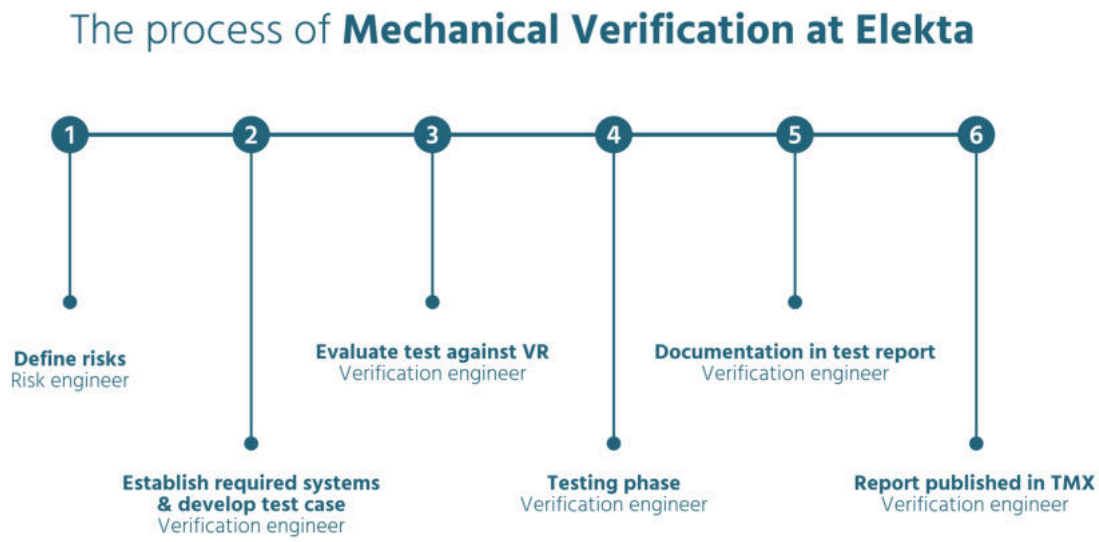


Figure 3.10: The process of Mechanical Verification at Elekta

The most commonly performed tests at Elekta are Lifetime tests, wear tests, precision tests, and displacement tests, summarized in bullet points below. The purpose of lifetime testing is to evaluate the likelihood of wear-out-induced failures during the product's lifetime. This evaluation aims to estimate the product's compliance with the long-term reliability requirements. [42] The objective of burn-in tests is to enable the detection of any problems with the product before being released to the market and customers [38]. Throughout the lifespan of products, various types of wear can occur due to handling and usage. Wear tests are employed to evaluate the impact of wear on components, materials, and surfaces. [9]. Precision tests are performed to assess the consistency of performance when a product undergoes repetitive testing, thereby indicating its reliability [32]. Elekta defines displacement tests as measurements that examine how products and systems are affected by patients' movements during treatment, equipment docking, or external factors. The obtained results are evaluated against specific tolerances to ensure the reliability of the systems.

Test-types:

- **Lifetime test**: Evaluate wear-out-induced failures during product lifetime
- **Burn-in test**: Detect problems before releasing product
- **Wear test**: Evaluate wear impacts
- **Precision test**: Assess consistency of performance, indicating reliability.

3.3.3 Tools

Various tools are utilized during the different testing procedures to accurately measure and simulate force, torque, and displacement. These tools are essential for conducting all the different tests, and evaluating the functionality and lifespan of products.

The most frequently utilized tools in testing procedures at Elekta include:

- Measurement robot, used for precise measurements
- Manual indicator (measuring watch), used instead of digital ones due to costs.
- The Allen keys
- Holding tools
- Torque tool

3.3.4 Efficiency

Efficiency is of great significance during Elekta's testing process. However, certain factors were identified that do not contribute to optimizing testing efficiency. One key advantage is the utilization of part-time workers who are students. By employing them specifically when test subjects arrive at Elekta, the company ensures a time-effective use of employee resources combined with and continuous testing process. In cases where students are unable to perform the tests, verification engineers at Elekta step in. However, this can be inefficient for their

primary work at the company. During certain verification tests that necessities the involvement of engineers, two professionals collaborate to operate the manual indicator. One engineer is responsible for conducting the measurements, while the second engineer handles the documentation of the results, see figure 3.11. It has been observed that this manual operation generates an inefficient test process and use of resources.

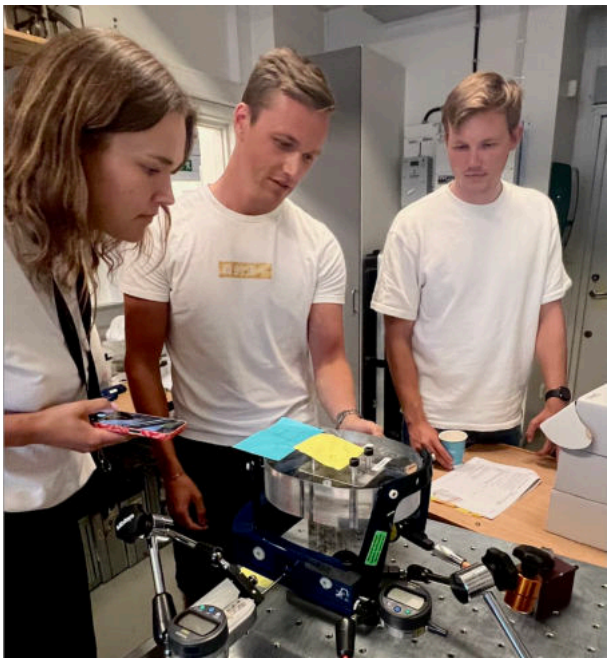


Figure 3.11: Elekta engineers demonstrating test on Leksell Vantage system

Certain tests, such as the washing machine test, involve significant waiting periods. While some testers attend to other tests during this time, others must wait until the machine completes its cycle. Additionally, some tests can be physically demanding and even painful, prompting testers to take breaks or work in pairs to substitute for each other. If a specific test has not been performed for a while, testers may need to relearn the procedures, potentially impacting overall efficiency.

3.3.5 Safety

Elekta has a safety briefing with all employees who are going to work in the lab. They offer gloves and complementary tools. The lab at Elekta presents some safety risks, firstly, the hot autoclave and dishwasher pose a potential danger of burns for the testers if proper precautions are not taken. Additionally, the presence of chemicals in a locker raises the risk of accidental contact, emphasizing the importance of safe handling procedures. The placement of the autoclave close to the chemical locker further compounds the safety concerns, as it increases the likelihood of potential mishaps. Moreover, working alone in the lab at night without surveillance poses a significant safety risk, as there is no immediate access to assistance or security in the event of an accident.

3.3.6 Ergonomics

The subject that received the most attention during the interviews was the matter of ergonomic concerns that emerged in combination with the testing procedures. The interviewees expressed discomfort and pain experienced while engaging in repetitive and monotonous movements. One of the Interviewees expressed their concern by stating, *"The wear and tear tests are terrible, I experience pain everywhere" - Interviewee 2.*

During the interviews, an interactive exercise involving mapping and discussion of ergonomic risks was conducted, the maps were combined resulting in the ergonomic body map presented in Figure 3.12

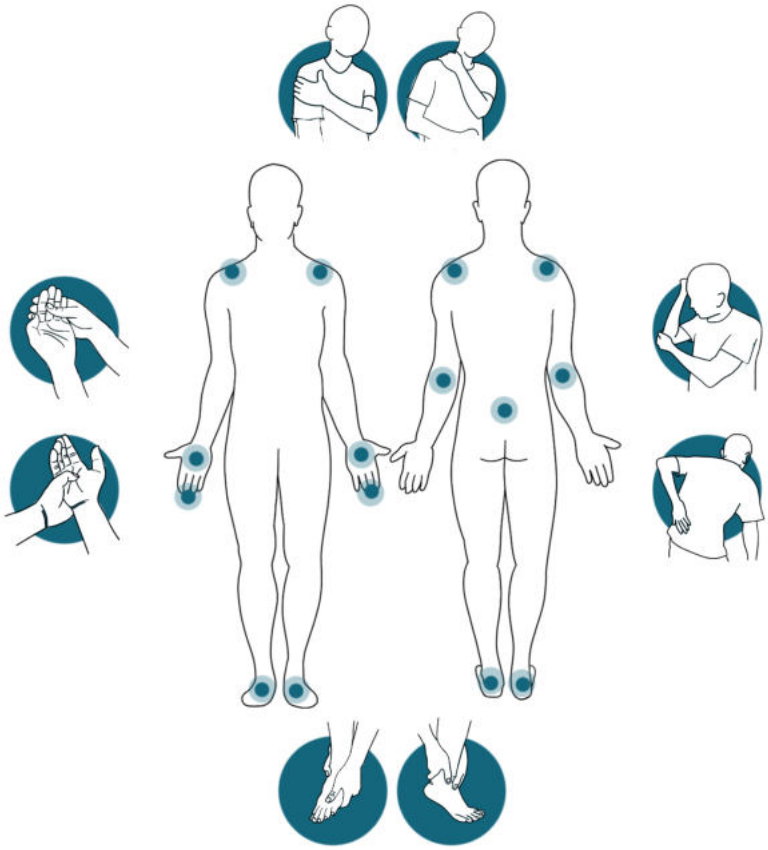


Figure 3.12: Ergonomic pain map

Interviewees 1-4 emphasized the discomfort in their hands and fingers while performing tests, stating *"The pain in hands and fingers is the worst" - Interviewee 3*, *"I get blisters and chafe on my fingers when performing certain lifetime tests" - interviewee 1* and *"The amount of pressure that I have to apply to inadequately designed load cells hurts the inside of my hands every time" - interviewee 4*. This observation aligns with the fact that all product verification tests are performed manually, highlighting hands and fingers as the primary source of ergonomic pain. During lifetime verification tests, testers repeatedly manipulate components using their fingers, enduring cycles of up to 10,000 repetitions. These tests require significant finger pressure against sharp edges and surfaces, resulting in soreness and tenderness in hands and fingers. The ergonomic issues in hands and fingers arise due to three risk factors of biomechanical exposure, repetition, force, and contact stress, which are described in section 3.1.9.

The majority of tests are conducted while standing. Due to the long duration of testing cycles, testers often experience foot pain from standing still for an extended period of time. This discomfort arises due to the risk factor of static posture described in section 3.1.9.

Several interviewees encountered soreness and pain in their shoulders, elbows, and lower back after repeatedly testing components connected to the bed, such as the real-time HD motion management fixture, see figure 3.13. Lifting a 2kg component in an awkward position for an extensive amount of cycles poses ergonomic risks. The severity of the pain forced the testers to alternate with each other during the test to manage the discomfort. The ergonomic issues that the operators encountered during this activity arose due to three risk factors of biomechanical exposure, repetition, force, and awkward position, all described in section 3.1.9.



Figure 3.13: Testing bed components

Ergonomic issues tend to manifest early on and intensify throughout the entirety of the test procedure. Interviewee 3 shared their personal experience stating *”I*

performed a test where I had to move a locking mechanism 1000 times in a day, the movement started to hurt after 50 times, so I continued 950 more times with pain" - Interviewee 3.

3.3.7 Limitations

While implementing a Cobot for testing purposes, several limitations need to be considered. Firstly, the ABB GoFa Cobot is unable to execute maneuvers requiring two hands, restricting the ability for complete automation of tests requiring these specific actions. For instance, it cannot operate the dishwasher or the autoclave due to their complex movements and the need for two-handed operations. Additionally, the Cobot is restricted from lifting more than 5 kg, thereby precluding the automation of tests involving a higher ergonomic risk factor due to increased force.

Additionally, when for example conducting tests on the arc system, it is essential to assemble components with varying heights to simulate different situations and test durability. This task becomes difficult with the Cobot due to its limitations in adapting to different configurations. Depending on the test, it can be more efficient to conduct it manually than to program the Cobot, since this also will require additional employee resources.

Lastly, if a test necessitates human touch, behavior, or intervention, the Cobot's capacity to simulate such motion may be restricted. The Cobot is designed to be able to imitate human movements. Through programming, it can perform irregular motions to simulate human behaviors. Hence, the extent of this limitation could potentially diminish with dedicated efforts toward developing irregular Cobot motions.

3.3.8 Interview ABB

This section presents insights gathered from an interview conducted with an industry expert employed at ABB. The interview aimed to explore the use of collaborative robots (Cobots) in different industries for testing and verification purposes. The insights provided valuable perspectives on the advantages and applications of Cobots in various scenarios. These insights were analyzed to extract meaningful findings, key findings are presented in Figure 3.14.

Cobots are used by Volvo to orbit around and scan various positions of an engine to verify the presence of all components. Equipped with a camera, the Cobots serve as complementary tools to human inspection, enabling the detection of details that humans might miss. Furthermore, Cobots are frequently used within the furniture industry. They are used to perform lifetime tests on furniture to test their durability throughout their life span. In these tests, Cobots apply pressure similar to that exerted by a human, simulating repetitive usage patterns over a product's lifetime. The interviewee also shared insights from a welder using a Cobot for assistance in various assignments. Upon receiving a welding assignment, such as the task of welding 200 flat iron pieces, the operator proceeded with creating a program by manually moving the Cobot through the desired path using the lead-through function. This approach allowed the operator to complete the welding task effectively. However, the operator chose not to save the program created during this process, due to the ease of generating a new program for future tasks. The key factors in this approach were simplification and ease of use.

When opting to simulate a motion including purely linear motion a Cobot with six degrees of freedom is not necessary to be used. However, the interviewee emphasized that when performing product verification tests, human movements often what to be simulated. These movements are not entirely linear, therefore the value of using robotics increases. Cobots enable more accurate simulations, making them worthwhile for such applications. Cobots can execute tests closely resembling human behavior. The simplicity of creating test conditions, utilizing wizard easy programming, and mimicking human actions are advantageous for

verification purposes.

Unlike traditional robots that require external safety measures and dedicated test rooms, Cobots are mobile and can be placed next to the product, saving both time and resources in implementation.

Regarding grip tools, it is according to the interviewee common to purchase a generic grip body and customize the fingers based on the specific application. Custom fingers can be easily interchanged between tests using screw connections or quick-release mechanisms. Recommended Cobot grip tool manufacturers include Zimmer, Schunk, Schmalz, and Piab.

The interviewee expressed belief in the future growth of Cobot utilization for validation, verification, and testing purposes. Cobots offer simplicity and efficiency, making them suitable for various applications, including repetitive tasks.

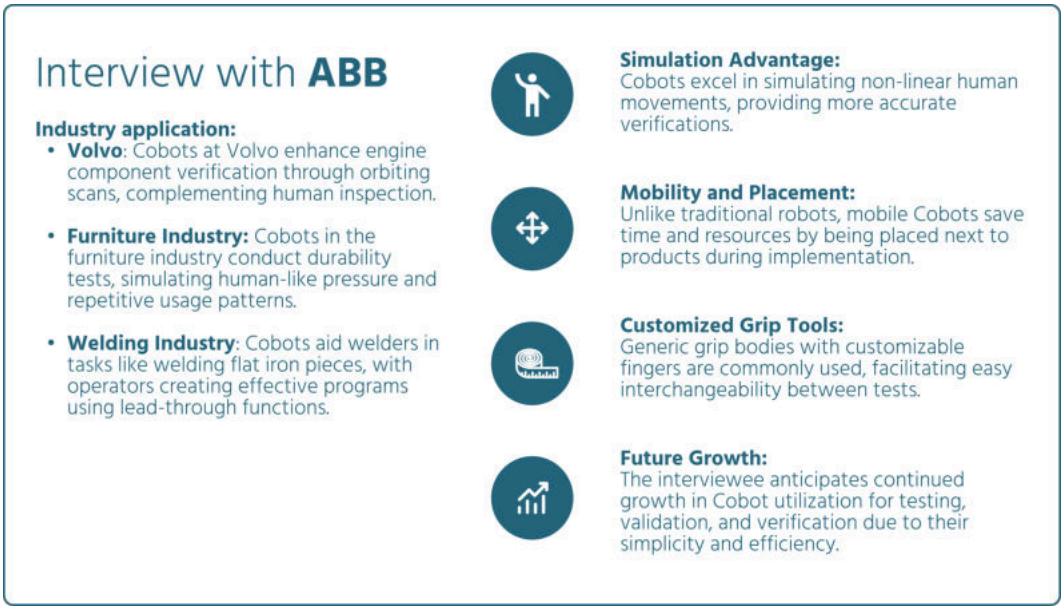


Figure 3.14: Interview ABB

3.3.9 Interviews Atlas Copco

Atlas Copco utilizes Cobots both in their production line and for the mechanical verification of their products. During a visit to their manufacturing site in Tierp, valuable insights were gathered through interviews and observations, key findings are presented in Figure 3.15. Atlas Copco continuously works towards automating operations to improve effectiveness. Their automation process started with a digitalization strategy, where potential areas of automation were identified. Following, the operations were determined as long-term or short-term, where activities with a short-term future of operation were eliminated for possible automation. After defining the areas of implementation an extensive pre-study was conducted to define the potential of automation. The prestudy contained research combined with testing that delivered information about processes suitable or not suitable for automation.

The parameters that were of great value for automating a process were first, the long-term aspect of the operation. For how long will the procedure be running and would automation be beneficial over that period of time? Secondly, they looked to if automation would improve the effectiveness of the operation. Third, the ergonomic aspects were prioritized where procedures caused injuries, problems, or discomfort.

Discussing the pros and cons of Cobot implementation with Atlas Copco resulted in further insight regarding the corporate effects of automation. Cobot implementation brings benefits such as standardized operations, improved quality, and results, combined with cost-effectiveness. However, it limits the flexibility of changing products when a system is built around the product's geometry.

A monotonous workload was the first and foremost reason behind ergonomic issues at Atlas Copco. A whole life of performing the same movements creates a series of ergonomic problems for operators. The most experienced aches and pains were located in the shoulders, wrists, and neck with an origin from

monotonous work where activities with a lot of force, repetition, or precision were required.

When testing and developing gripper attachments the main focus was to create an interface that could handle the product without damaging it. To create this interface resources were put into the geometry of the gripper. When modeling the gripper most insights were gained when testing different gripper variations to generate the most efficient, ergonomic, and simple solution. *"After developing numerous complex solutions, we realized that a simple solution is often the best solution" - interviewee 6.* With a simple solution, interviewee 6 argues that fewer complications follow, such as wear or the need for service. In the search for a gripper solution that would not damage the product, employees at Atlas Copco experimented with 3D printing solutions in numerous different materials. Resources were also put into exploring different grippers, variations in clamping force, static locking mechanisms, and number of fingers and joints to comply with the product's interface. Furthermore, the aim was to make the operation as easy as possible for the operator. *"It should be easy to do the right thing" - interviewee 6.* Given their utilization of a collaborative robot, it is essential to emphasize not only the Cobot itself but also the human-robot interface.

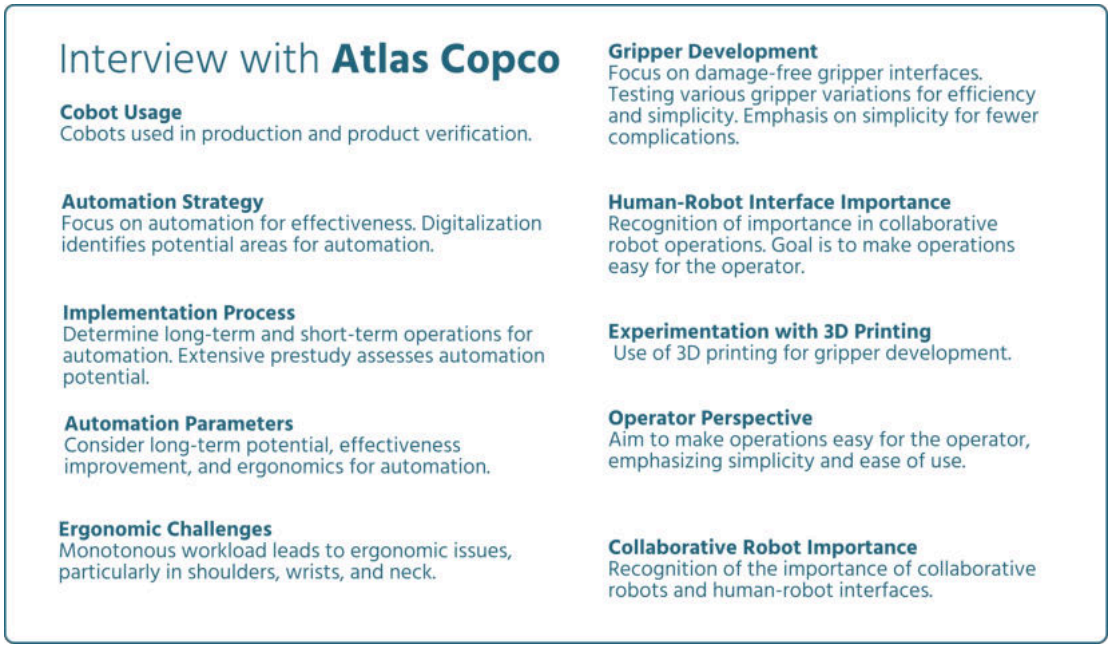


Figure 3.15: Interview Atlas Copco

3.3.10 Observations

A series of observations were carried out with the interviewees at Elekta and Atlas Copco to assess movements, effectiveness, and performance during testing, and also to comprehend the human-robot interface. Observations were conducted with three different products at Elekta. Firstly, the observation focused on lifetime testing of the real-time HD motion management fixture, where interviewee 3 demonstrated the testing procedure of rotating the fixture arm while both standing and sitting. When demonstrating this testing procedure an awkward posture of the operator was identified where the tester had to rotate his arm 180 degrees with high force calling for movements with repeated rotation of the flexed shoulder, Illustrated in figure 3.13

The second observed procedure involved a lifetime test of the Vantage frame. Where monotonous movements were executed during an extensive number of cycles. The test required operator resources to perform the simple repetitive movement, which resulted in inefficient resource management.

The third test involved rotating a metal cylinder to test its lifetime wear. During

the test, the operator grips the cylinder with his fingers and twists his wrist 1000 times. As described in section 3.1.9 a pinch grip, such as this one, produces 3-5 times more force on the tendons in the grip and hand, resulting in possible ergonomic injuries when performing this test for extensive cycles of repetition. Interviewee 3 complains of thumb pain from the firm grip and released metal shavings that irritate the fingers.

At Atlas Copco, observations were carried out analyzing their Cobot system. The operator had little involvement in the system eliminating their previous ergonomic issues. The Cobot was manufacturing needle bearings, which is an operation that requires high precision in a static posture for a human operator. With Cobot execution the operator simply had to refill the needles handled by the Cobot and start the program. The results from automation were increased effectiveness, higher precision, and quality combined with better operator health.

4 Analysis

This chapter focuses on the analysis, evaluation, and definition of knowledge obtained through a comprehensive exploration of both theoretical and market research. The aim is to identify key factors that influence the selection of a gripper for verifying and testing Elekta's products and to assess the suitability of Cobots for product verification processes. Additionally, the chapter examines valuable insights gained from user research. By exploring these topics, the chapter provides readers with the necessary knowledge and insights to design and evaluate grippers effectively, leading to enhanced productivity and efficiency in automated testing and verification processes.

4.1 Insights from theoretical and market research

This chapter focuses on the analysis, evaluation, and definition of the knowledge acquired from both theoretical research and market research.

4.1.1 Automation

Analyzing the theoretical research provides valuable insights into the significance of automation and human-robot collaboration, particularly in the context of Industry 4.0. Industry 4.0, as the fourth industrial revolution, emphasizes the integration of robotics, automation, artificial intelligence, and the Internet of Things. The primary goal is to enhance productivity and product quality. Elekta, being at the forefront of medical technology, can leverage these advancements to improve the efficiency, quality, and flexibility of its products and services.

Incorporating Cobots into Elekta's operations offers many advantages. Firstly, the Cobot enhances efficiency by handling repetitive tasks, allowing human operators to focus on more creative, critical, and decision-making-intensive aspects of their work. Secondly, the integration contributes to heightened product quality in Elekta's medical technology by combining human judgment with robotic precision. Thirdly, a streamlined collaboration between humans and Cobots results in time savings. Additionally, both ergonomic and safety risks can be reduced and create a secure and healthier work environment, reducing the

risk of injuries for Elekta's employees. Adopting Cobots aligns with Elekta's dedication to innovation and excellence in the medical technology sector, marking a valuable step in adapting to industry changes and utilizing the benefits of Industry 4.0.

4.1.2 Gripper evaluation

When selecting a gripper for verifying medical products, several factors need to be considered. The object's size, shape, weight, material, and required precision play a significant role in choosing the right gripper. All grippers mentioned in the report have their strengths and limitations, see figure 4.1

Gripper type	Function	Strengths	Limitations
Finger gripper or parallel gripper	Mechanical gripper, Electric or pneumatic driven, 2-4 fingers	Ability to grasp most shapes, high grip strength, firm and secure grip	Difficulties gripping slippery surfaces, lift heavy objects, can damage sensitive materials
Vacuum gripper	Air pressure	Cheap gripper, secure attachment, can manipulate very large and heavy objects, can handle diverse materials and surfaces	Expensive in operations, not suitable for small components and irregular shapes, the surface must be smooth, non-porous and flat
Soft gripper	Delicate silicone or plastic grip	Handles fragile objects, non-destructive and gentle gripping, suitable for handling complex geometries	Low grip strength, less secure grip, cannot handle heavy objects
Magnetic gripper	Magnetic force	Strong reliable grip, no risk of damage or deformation, straightforward and rapid gripping	Not suitable for ferromagnetic materials or uneven/curved shapes, weight limitation, objects can lose grip, affected by oil
Gecko gripper	Adhesive technology	Can adhere to diverse surfaces, non-destructive/gentle gripping, cost-effective, reliable for handling small components, can handle different surfaces	Can not conform to curved surfaces

Figure 4.1: Gripper strengths and limitations

4.1.3 Elimination matrix

A matrix of criteria values was created to assist Elekta in selecting an appropriate gripper based on the specific requirements of their tests. The matrix incorporates criteria such as the affordability of the gripper, grip strength, and its capability to grasp various surfaces and shapes. Additionally, considerations for safety measures, gripper weight, product imprint, and potential damage risk were taken into account. The matrix also includes factors like precision, customization, object

size and weight, ability to handle dirt, and low operational costs. For further details, see appendix A.2.

The matrix considers and grades the gripper's strengths and weaknesses based on the information gained from the research chapter see 3.2.2. Furthermore, with the help of Elekta's testing experts, their different tests, lifetime/wear, precision, and displacement were graded depending on the importance of the criteria. Although every criterion has significance, this matrix serves as a guiding tool for Elekta in future scenarios where they may encounter distinct test cases with specific and crucial criteria.

For medical product verification, the RG6 finger gripper from OnRobot chosen by Elekta stands out as a versatile and efficient tool. In the matrix, see appendix A.2 it displays high grades of important criteria such as strong gripping force, ability to grip different shapes, and smart sensor technology which makes for secure and reliable gripping. Furthermore, it has high precision and customizable fingers that cater to various objects, making it a valuable choice for automating repetitive tasks. In the different test cases, this gripper proves to overall be a preferred choice.

Overall, the evaluation of grippers for product verification involves carefully considering the specific requirements and characteristics of the objects being handled. The chosen gripper should provide a secure and precise grip, while also offering adaptability, integration capabilities, and compatibility with the Cobot system. Further development to be more adaptable to the products would be to customize the fingers according to the test objects and simulate human touch to simulate real-life testing. Additionally, two hand grippers would be beneficial to be able to execute two hand movements.

4.1.4 Key test parameters for Cobot implementation

Upon researching the capacity of the Cobot and the OnRobot RG6 Finger Gripper, key parameters were identified navigating the process of choosing suitable product verification tests that could be performed by a Cobot. The list of key parameters in table 4.1 serves as a framework against which tests can be evaluated to confirm a suitable Cobot implementation. To clarify the table 4.1, the maximum payload indicates the overall load capacity of the Cobot and gripper, while the gripping force refers to its ability to securely hold objects. This holds significant importance in the context of Elekta's products, ensuring that they are not dropped or subjected to defects. The test reach corresponds to the farthest distance the gripper can operate, and the product length indicates the maximum extension of the gripper jaw, enabling it to securely grasp parts of a specific size. The gripper is particularly well-suited for regularly shaped objects, providing optimal and secure gripping. Furthermore, it is the most effective in lifetime tests, which often involve monotonous and repetitive tasks.

Table 4.1: Key test parameters for Cobot implementation

Key test parameters for Cobot implementation	
Maximum payload	3.75 kg
Maximum test reach	950 mm
Gripping force span	25 - 120 N
Maximum product length	160 mm
Optimal product geometry	Regularly shaped
Optimal test duration	Lifetime
Optimal test type	Repetitive

4.2 Insights from user research

Analyzing and clustering the information gathered during the user research phase resulted in numerous valuable insights. The primary drivers for Cobot implementation in verification processes were to eliminate ergonomic issues for tester personnel, combined with the desire to enhance the efficiency of tests

with extended cycles. Ergonomic problems commonly arose when personnel conducted lifetime tests involving repetitive and monotonous movements. As these tasks can be performed by a Cobot, its implementation can potentially increase testing efficiency while also improving the health and well-being of test personnel. The implementation of the Cobot can also play a vital role in enhancing laboratory safety, particularly during high-risk working hours, such as night operations, by effectively replacing employees working alone. However, the main hindrance to Cobot implementation lies in the current limitations of its capabilities. Certain tests necessitate two-hand operations, which Cobots cannot currently fulfill. The main insights are presented in a user analysis 4.2, and further elaborated in the following paragraphs.

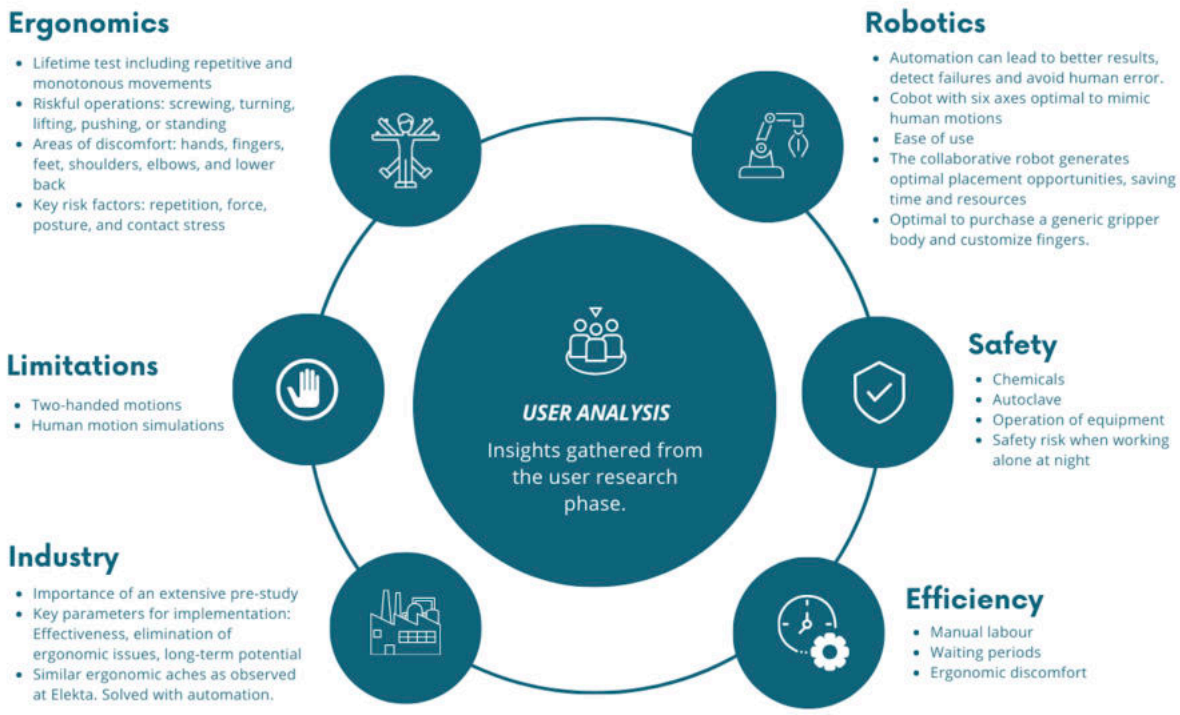


Figure 4.2: User analysis

Efficiency: Currently, mechanical verification tests are carried out manually. This manual approach has proven to, in some cases, lead to inefficient work and unnecessary employee resources. Certain procedures necessitate significant waiting periods, either to address ergonomic discomfort or to wait for processes, like the completion of an autoclave procedure.

Safety: Safety in the laboratory is closely related to the proper use and storage of chemicals, the handling of the autoclave, and the correct operation of equipment. However, a significant safety concern arises when working alone at night in the laboratory without supervision. Employee safety is at risk since there is no immediate access to assistance in the event of an accident or emergency. To mitigate this safety risk Cobot can replace human presence during high-risk working hours, reducing the safety risk associated with working alone.

Ergonomics: Analyzing the ergonomic problems encountered during the testing processes, several common factors have been identified. Testers consistently experience pain or discomfort when conducting lifetime tests involving repetitive and monotonous movements for an extensive period of time. Pain is commonly experienced when engaging in activities such as screwing, turning, lifting, pushing, or standing for extended periods, especially during a large number of cycles. The pain or discomfort frequently manifests in the operator's hands, fingers, feet, shoulders, elbows, and lower back. This pain arises due to biomechanical exposure, where repetition, force, posture, and contact stress are primary risk factors.

Limitations: Using a Cobot for mechanical verification brings forth some limitations. The Cobot cannot operate two-hand motions, this eliminates certain tests for being automated. The six-axis Cobot can simulate human motion, but the similarity can still be limited. Further limitations gathered during the theoretical research are presented in section 4.1.4.

Robotics: Insight from discussions with experts within the field of Cobots resulted in valuable insights. A Cobot has better precision than humans, a switch from manual to automatic labor can lead to better results, failures can be detected and human errors can be avoided. A Cobot with six axes is optimal to mimic human motions, generating the ability to simulate human usage during a product's lifetime. The Cobot's simplification and ease och use enable broad

utilization opportunities within different industries. Collaborative robots can be placed next to products and humans, saving time and resources during implementation. The recommended use of grippers was to purchase a generic body and customize fingers based on specific applications.

Industry: During the interviews conducted with experts at Atlas Copco, valuable insights were gained regarding the role of Cobots in the industry. One key finding emphasized the significance and value of conducting an extensive pre-study to define the potential of automation within a particular industry. The pre-study serves as a foundation for informed decision-making and the successful development of robotic attachments. The experts identified certain parameters that are important to consider when implementing a Cobot in an industrial context. These parameters contain the improvement of effectiveness, the elimination of ergonomic issues, and the assessment of the process's long-term potential. The noticeable improvements from Cobot usage were the potential of standardized operations combined with improved quality and results. Furthermore, the operators at Atlas Copco have encountered ergonomic challenges while performing their manual work within the industry. These ergonomic issues arise from engaging in monotonous tasks that require high precision. As a result of these activities, operators have experienced discomfort and pain in their shoulders, wrists, and neck areas. The pain experienced in these regions is similar to the discomfort observed at Elekta, with a foundation in biomechanical exposure, where repetition, force, posture, and precision are primary risk factors. This confirms the sources of ergonomic discomfort and pain and their potential for improvement through automation.

4.3 Test definition

Elekta employs a comprehensive range of particular tests, aimed at diverse objectives such as lifetime, wear, precision, and displacement, among others. The selection of the appropriate test for this project was governed by three key pillars: ergonomic considerations, efficiency, and the level of difficulty involved.

Ergonomic concerns predominantly arose from the lifetime and wear tests, where long exposure to repetitive and monotonous movements was observed. These tests involve multiple cycles and put a considerable strain on the testers, also impacting the overall efficiency. Moreover, the assessment acknowledged the influence of the human factor, where longer task duration, potential injuries, and the need for breaks or task alternation contributed to reduced efficiency. Safety hazards associated with ergonomic factors and lab protocols were also factored into the evaluation. Efficiency emerged as a critical aspect, considering the impact of task performance by humans. Both ergonomic and lab safety concerns further underscored the significance of an efficient approach.

The final pillar, the degree of difficulty in the test, was also assessed. Importance was given to tests requiring complex movements and challenging problem-solving capabilities. Based on these pillars, it was apparent that lifetime and wear tests held significant promise as the developmental focus for this project. To complement this, a Cobot was considered suitable to be implemented for certain tests. Elekta's test experts actively participated in discussions related to test selection, offering valuable insights. Notably, they presented a particular test scenario where they desired to optimize efficiency and minimize ergonomic injuries.

Henceforth, with the help of the three key pillars and Elekta's recommendation the Leksell Vantage CT Fiducial Box (CT box) and frame holder emerged as the chosen product for this project, see figure 4.3.

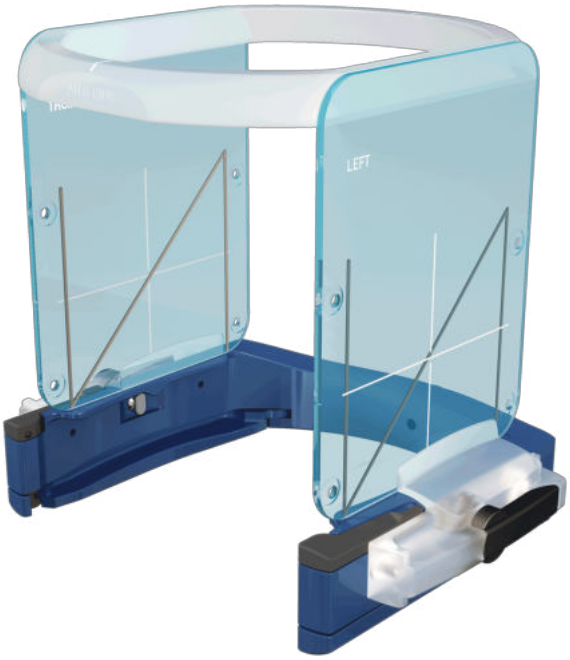


Figure 4.3: Leksell Vantage CT Fiducial Box and frame holder

4.3.1 Leksell Vantage CT Fiducial Box and frame holder

The Leksell Vantage stereotactic system, see figure 4.4 is used in radiation therapy, specifically in stereotactic radiosurgery and stereotactic radiotherapy procedures. It is designed to be used in conjunction with Elekta's Leksell Gamma Knife radiosurgery system.



Figure 4.4: Leksell Vantage Stereotactic system

There are several tests conducted on that system, one of the tests evaluated in this report contains two parts, the CT box and a frame holder. Both the CT box and the frame holder play a crucial role in the planning and delivery of the radiation treatment when using the gamma knife.

The CT box holds fiducial markers attached to a patient's head during imaging processes, such as CT and MRI scans. Fiducial markers act as reference points for precise alignment in radiation treatment. The frame holder secures the Leksell stereotactic frame to the head, ensuring stable positioning. This rigid frame aids accurate targeting of radiation in the brain. Together with the CT box, they help localize the target and deliver radiation accurately, minimizing damage to healthy tissues. The product is compatible with diagnostic CT scanners via Leksell CT Table interfaces or intraoperative CT scanners via OR table interfaces. The stereotactic CT box is docked to the frame holder to hold the head frame in a stable position.[26]

4.3.2 Lifetime test with the CT box and frame

The selected lifetime test for evaluation contains the Leksell Vantage CT Fiducial Box and Frame Holder, focusing on evaluating the product's long-term durability. This test is particularly significant, as the CT box regularly undergoes removal and re-insertion from the frame holder in real-world hospital scenarios, a crucial step carried out by medical staff before radiation treatment. The test procedure entails a series of steps designed to simulate real-life usage conditions and investigate the product's ability to withstand wear and tear over its lifetime.

These steps are as follows:

Step 1: The two black levers on each side of the CT box are pulled upwards. Subsequently, both sides of the CT box are flexed outwards and lifted upwards, enabling its removal from the frame holder, see step 1 in Figure 4.5.

Step 2: The CT box is then flexed out again and docked from above onto the frame holder. Special attention is given to ensuring that it securely clicks into the brackets marked with white indicators on the opposite side of the frame holder, see step 2 in Figure 4.5.

Step 3: Finally, the two levers on each side are turned down to securely lock the CT box in place within the frame holder, see step 3 in Figure 4.5.

This process is repeated 500 times to thoroughly evaluate the product's components and materials for potential wear and tear during its operational lifespan. Given the critical role of this product in ensuring precise and accurate treatment delivery, this durability testing is of vital importance in validating its long-term performance in real-world medical applications.

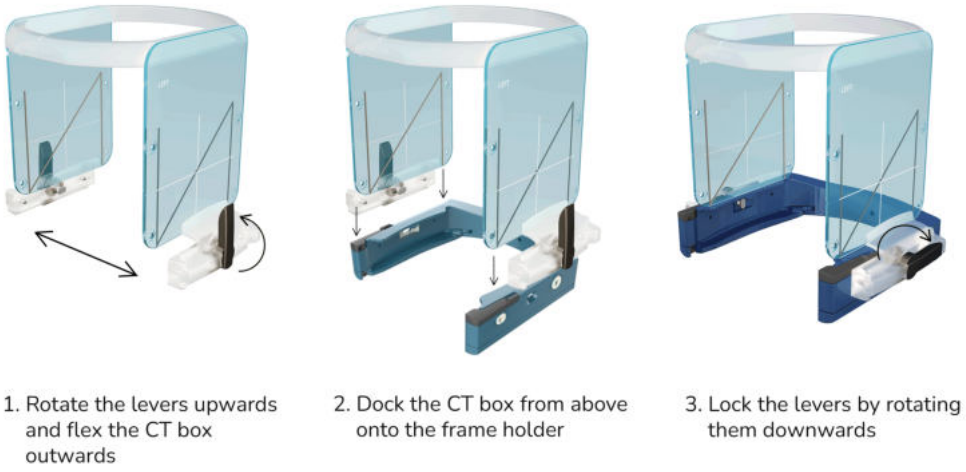


Figure 4.5: Leksell Vantage CT Fiducial Box and frame holder test in steps

5 Creative process

This chapter is centered around the development phase, which is structured based on the selected verification test of the CT Box. The phase commences with an evaluation of the CT Box test to explore potential solutions in line with previously gathered insights. Subsequently, this is an iterative process of brainstorming sessions to generate concepts, followed by CAD modeling and 3D printing. The concepts are then tested and verified with the Cobot. This iterative process aims to continuously enhance each concept until a viable solution aligning with identified insights and requirements is achieved. In the final steps of this phase, a Cobot programming code is developed for the entire test, culminating in a demonstration of the final concept and program.

5.1 Evaluation of test execution

This phase was initiated with an analysis of the test execution, necessary to commence the development of the Cobot. The primary focus was on three critical aspects: evaluation of movement, gripping technique, and analysis of forces and torque acting on the product. These investigations served as the foundation for identifying potential solution paths for the Cobot and the following development of an appropriate gripper.

5.1.1 Evaluation of Movement and gripping technique

The initial step was a thorough examination of the movements employed during the testing process. The tester's actions were carefully observed, with particular attention given to twisting, pulling, and lifting operations. This process primarily entailed arm and hand manipulations, with the tester standing upright. Key observations revealed the necessity for a strong force to rotate the lever upwards, followed by a firm grip to flex the sides of the CT box for its removal from the frame holder. The same gripping technique was then employed to dock the CT box back into the frame holder, which occasionally required some system jerking to secure it in place. Finally, the lever was rotated down using the same force downward

to lock the CT box securely. It was noted that the entire process relied on a two-handed motion, making the removal and reinsertion of the CT box challenging and time-consuming.

5.2 Mechanism Exploration

To continue the evaluation, a brainstorming session and crazy 8s were conducted to explore potential mechanisms that could facilitate the required movements. One approach involved programming the existing finger gripper on the Cobot to pull up both levers simultaneously. However, achieving the two-handed motion on a one-armed Cobot necessitated a mechanism capable of flexing the CT box during both removal and reinsertion operations.

The brainstorming sessions resulted in several concepts of mechanisms developed to solve the issue of a two-armed movement, see figure 5.1. Three of the most promising ideas are presented below with simplified figures.

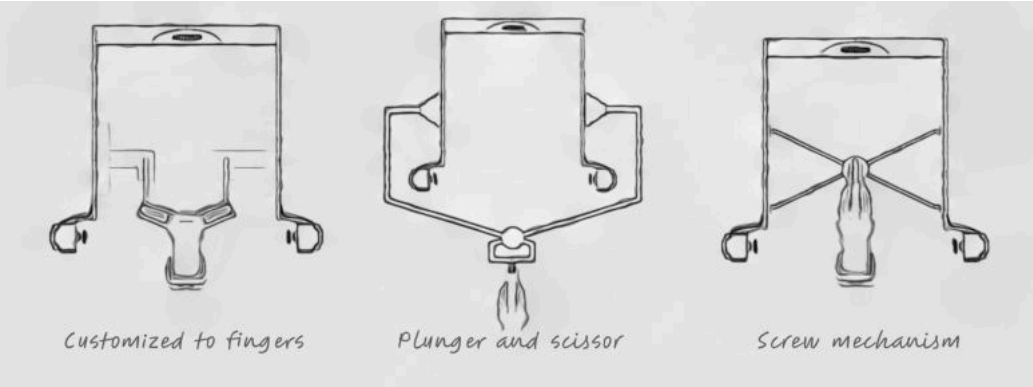


Figure 5.1: Sketches from the brainstorming session

Customized to fingers

This concept relies on customizing the gripper's fingers to carefully designed parts. The proposed approach involves utilizing the positioning holes located on the sides of the CT box to allow outward flexing on each side when the gripper is extending its jaw, see figure 5.2.

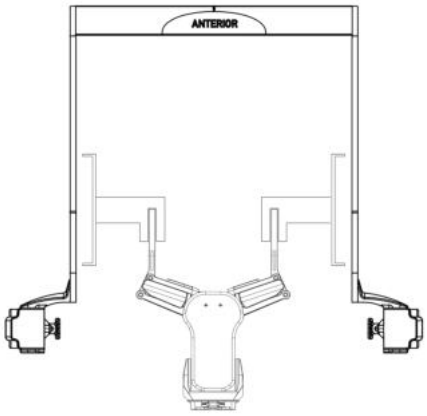


Figure 5.2: Sketch illustrating the customized concept for flexing CT box

Plunger and scissor mechanism

This concept employs suction cups affixed to the exterior of the CT box. Subsequently, a scissor mechanism is integrated into the plugs which allows for controlled inward and outward flexing by the use of a gripper. This enables manipulation from both above and below the CT box, see figure 5.3.

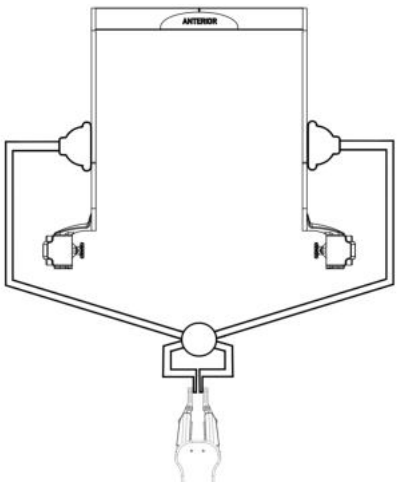


Figure 5.3: Sketch illustrating the plunger scissor concept for flexing CT box

Screw mechanism

The proposed screw concept involves a rotating wheel equipped with arms securely attached to the CT box. By rotating this wheel, the arms extend outward, facilitating the detachment of the CT box from the frame holder, see figure 5.4.

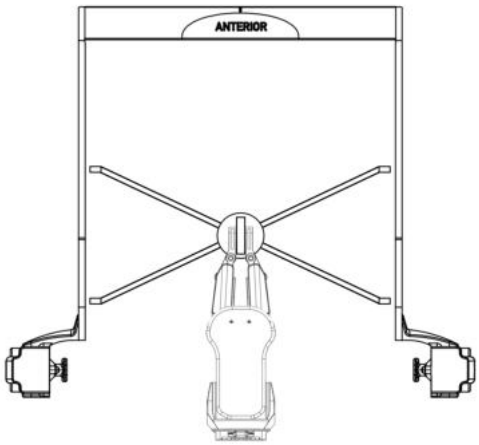


Figure 5.4: Sketch illustrating the Screw mechanism concept for flexing CT box

5.3 Different ways of docking the CT box

Upon undertaking the development of various concepts for the two-handed movement, it became evident that their complexity was more substantial than initially anticipated. This particular option demanded capabilities and tools beyond the robot's capabilities for this project. During a thorough examination of the box, an intriguing question emerged regarding the possibility of conducting the test one side at a time. To explore this possibility, discussions were initiated with one of the creators of the Vantage system, aiming to explore alternative approaches. Subsequently, it was revealed that conducting the test one side at a time was indeed possible. Doing so would necessitate exerting twice the force to achieve the same elongation in the box, as opposed to pulling both sides simultaneously. This information clarified the potential approach to be taken.

Before proceeding with further decisions, tests were carried out to compare the wear and efficiency of pulling one side versus both sides simultaneously. These analyses would serve as a basis for making an informed and strategic choice.

5.3.1 Forces and torque analysis

The dynamic behavior of the system during the test is influenced by both forces and torques during its movements, this is displayed in Figure 5.5.

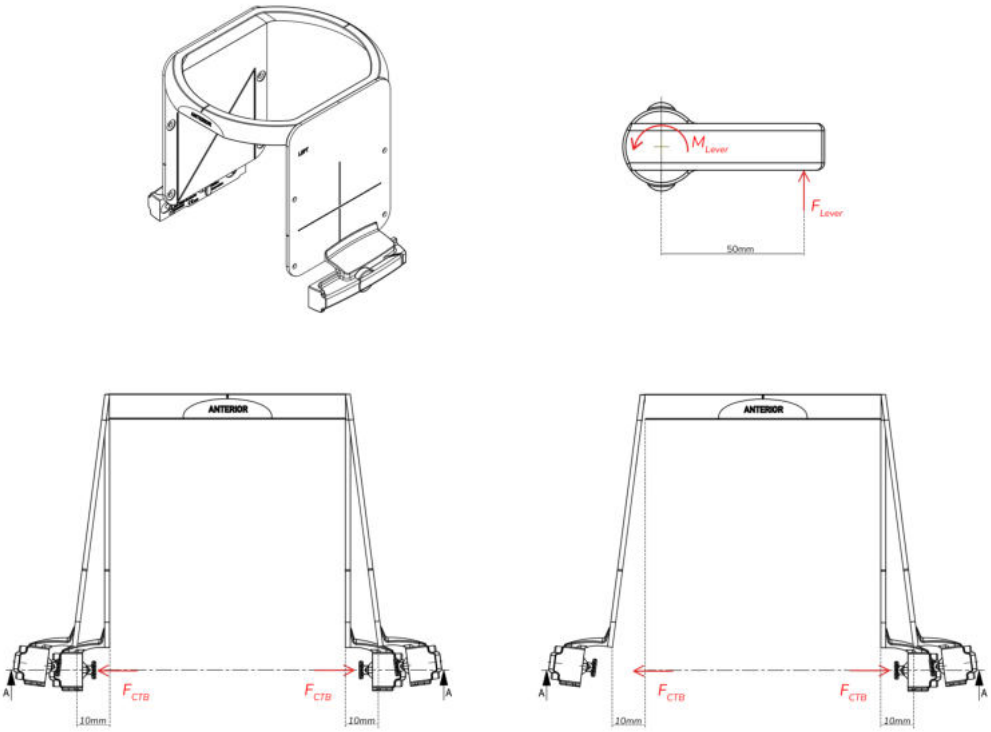


Figure 5.5: Leksell Vantage CT Fiducial Box and frame holder test in steps

The mass values for both the CT box and the lever arm were extracted from the CAD models, see values in table 5.1.

The force required to act on the CT box and lever is assessed using a dynamometer. The CT box force is measured until a deflection of 200mm is achieved, see table 5.1. While the lever force is measured until it undergoes a 90-degree rotation from

Calculation parameters for CT Fiducial box and Frame holder	
m_{Lever}	0,2 kg
F_{Lever}	20 N
m_{CTB}	0,7 kg
$F_{CTB,1side}$	15 N
$F_{CTB,2sides}$	10 N
$\delta_{CTB,1side}$	20mm
$\delta_{CTB,2sides}$	10mm

Table 5.1: Calculation parameters for CT Fiducial box and Frame holder

its initial position, see table 5.1

The initial rotation of the levers induces forces that subsequently generate a torque upon their attachment. When determining the torque, the length parameter is selected to be the maximum distance at which the robot can securely grip the levers. Additionally, the primary goal is to minimize the necessary force to the utmost extent, thereby enabling the Cobot to conduct the movement efficiently, see Equation 1.

$$M_{Lever} = F_{Lever} \times L_{Lever} = 1Nm \tag{1}$$

An investigation was conducted to ascertain whether a 20mm distance would generate an equivalent impact on the CT box when compared to the scenario where both levers are pulled. A positive outcome would confirm the feasibility of conducting the test on a single side, while still achieving the same effect.

As shown in 5.5, the CT box experiences equal forces whether flexed on both sides simultaneously or one side at a time. This phenomenon occurs because, for instance, when the right side is pulled outward, the left side remains securely docked to the frame. In 5.5, it is evident that stretching only the right side to twice its length produces an identical effect on the box as pulling both sides

simultaneously, see equation 2. This confirms that the test can be conducted on one side at a time and continues the exploration of new innovative approaches to accomplish the testing with the Cobot.

$$Xoneside = 2 \times 10mm = 20mm \tag{2}$$

5.4 Cobot gripper finger development

The phase began with a brainstorming session to develop appropriate fingers to be attached to the generic gripper body. During this session, various techniques for attachment to the CT box were generated, accompanied by a number of ideas that provided support and elaboration for each technique.

To push past the ideas generated during the initial brainstorming session, the creative method known as ”Crazy 8” was performed. This method challenged a creative generation of ideas, with the primary objective of creating a diverse and extensive range of solutions.

Once the idea bank had been filled with a wide collection of solutions, it allowed for the testing of ideas to validate their potential against the specific requirements of the Cobot and the corresponding test case. Each idea was constructed in CAD and 3D printed in Nylon material. Additional material was added to some mock-ups to increase friction between the finger and the CT box. Figure 5.6 presents the iterative process encompassing the various ideas and their development. Starting from the extensive idea bank and finishing with three further developed concepts. The following paragraphs will present the insights and evaluations conducted throughout this iterative process.

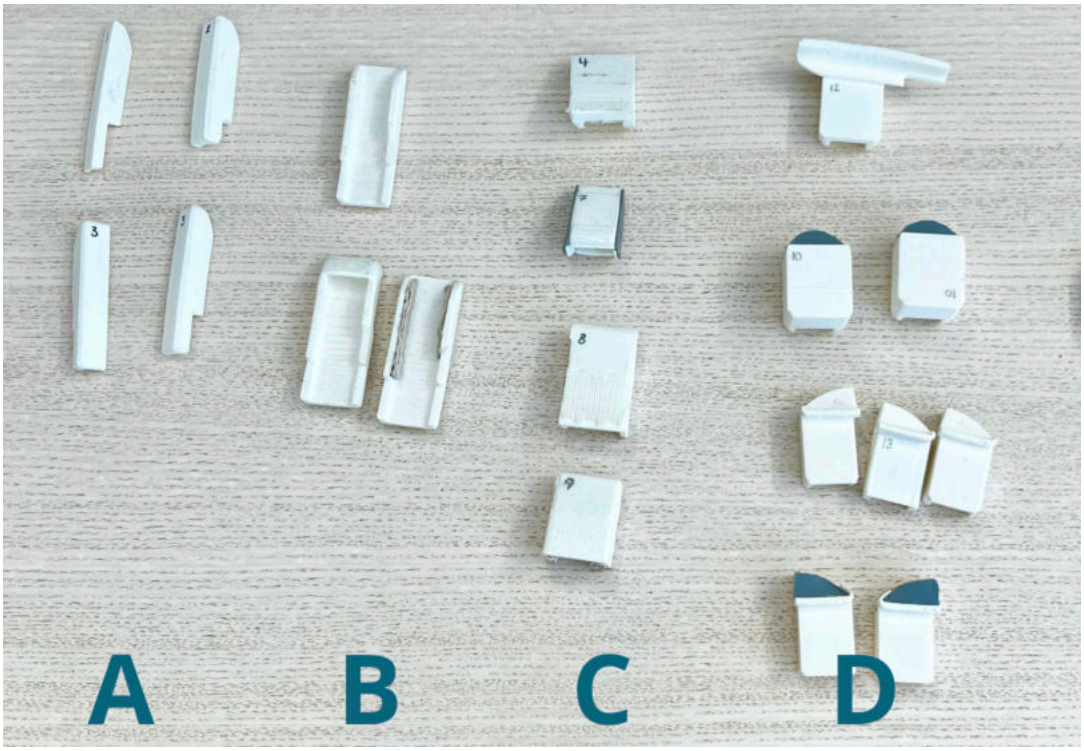


Figure 5.6: Idea development

The concepts presented in section A of the iterative process contain models that can be attached to the existing Cobot fingers to manipulate the locking levers by applying force from each side of the lever, see figure 5.7. Upon attaching these prototypes to the Cobot and assessing their performance, it was observed that the thickness of some models was inadequate to withstand the forces applied by the Cobot, while others were too bulky and collided with the frame, hindering the lever from locking completely. Consequently, these solutions do not offer an efficient procedure, as the operator needs to switch between four different components to conduct the test on both sides, leading to unnecessary human interaction and pause time.

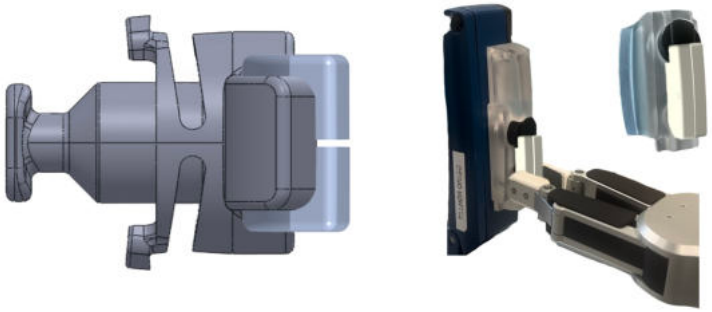


Figure 5.7: Showing idea area A on the lever and in use

The concepts generated in section B represent further developments of the initial ideas. These ideas were developed to overcome the inefficiencies observed in section A. In section B, the opposite sides were combined into a unified body, see figure 5.8 simplifying the test procedure by reducing the number of components to switch from four to two.

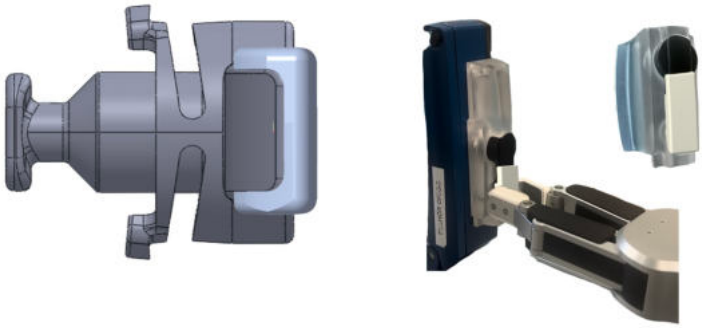


Figure 5.8: Showing idea area B on the lever and in use

By programming the Cobot to utilize these newly developed fingers during the test, additional insights were gained. During testing, it was discovered that the fingers tended to slide off the locking lever, and they were unable to flex the frame. To address this, friction material in latex was added to the designs. This material prevented the fingers from slipping off the lever and enabled them to flex the frame up to half of the desired flex distance before sliding off. Another challenge with the fingers in section B was that they were easy to attach incorrectly, lacking intuitive guidance for the operator. To tackle this challenge, a new finger was designed with a guiding finger imprint and a top lid, to secure the lever to the finger in an intuitive way during testing, see figure 5.9. Despite these improvements, testing

the final iteration of section B revealed a collision between the frame and the top lid, presenting new obstacles to address.

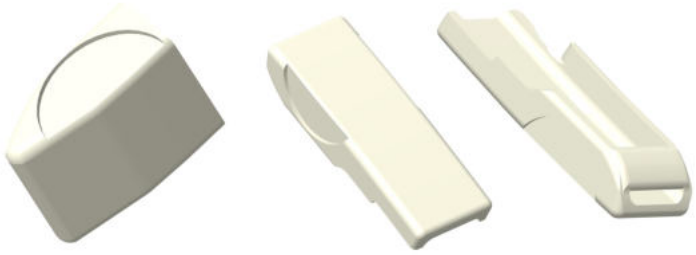


Figure 5.9: Finger design with a guiding finger imprint

In section C, the focus shifted toward exploring further improvements on how to securely grip the lever to prevent slippage. Numerous iterations were conducted, incorporating Cobot finger guidance and inner friction material into the designs, see figure 5.10. Various sizes were tested to generate a concept that accommodates the inner friction material while avoiding collision with the frame. However, the outcome revealed that this combination was not possible since it either got too bulky or fragile. As previously noted, a bulky design would risk collision with the frame, potentially leading to damage to the product. Conversely, if the design used was too fragile, it would fail to withstand the demands of all test cycles. Striking the right balance between design and material strength is essential to ensure both product integrity and the successful completion of the testing procedure. Furthermore, another challenge identified was the presence of two potential areas of slippage - one against the lever and the other against the Cobot fingers.

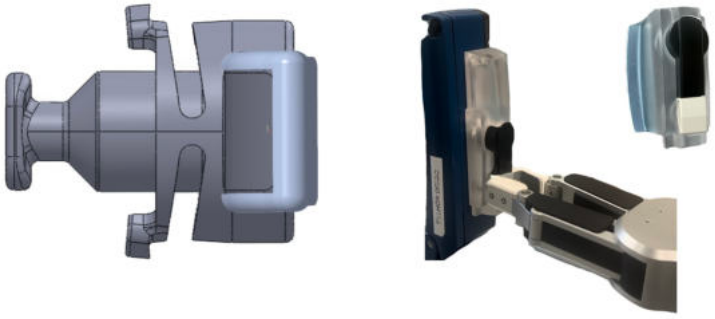


Figure 5.10: Showing idea area C on the lever and in use

Iterations in section D were based on the previous insight to scale down the areas of slippage. Various finger shapes were explored to test their capability to flex the frame and produce an intuitive user experience, see figure 5.11. To enhance the grip between the lever and finger, friction materials were incorporated. Despite these additions, the fingers still faced difficulties in gripping the lever with enough force to achieve the desired 2 cm of frame flexion. To further investigate, different gripping techniques were tested to safely grip and flex the product.

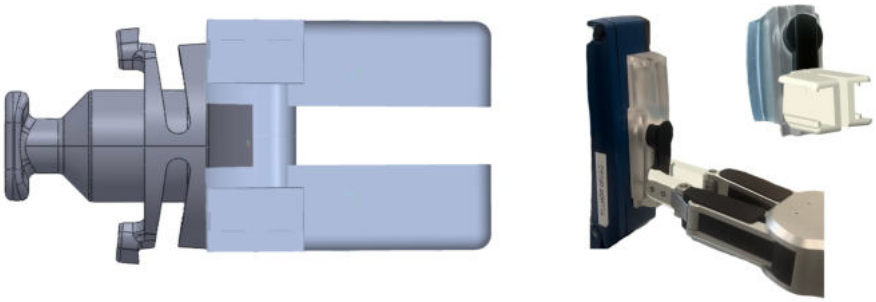


Figure 5.11: Showing idea area D on the lever and in use

The tests revealed that the friction between the materials was insufficient to facilitate the intended movement. Executing it in this way necessitates the use of two materials with highly textured surfaces. Even if these materials would function, there was no guarantee that they would deliver optimal performance for the large cycles of 1000 flexes on each side of the ct box, without possibly damaging the product. Therefore, the idea of using friction for flexing the frame

was eliminated.

With the decision to eliminate ideas using friction, combined with the insights to deliver a solution constructed as Cobot fingers which easily communicate with the user and does not necessitate a switch of fingers, new ideas started to take form, see figure 5.12. The importance of not damaging the product war bigger than the factor of effectiveness. Therefore, the decision was made to switch the grip between rotating the lever and flexing the frame. This generated the final concept.

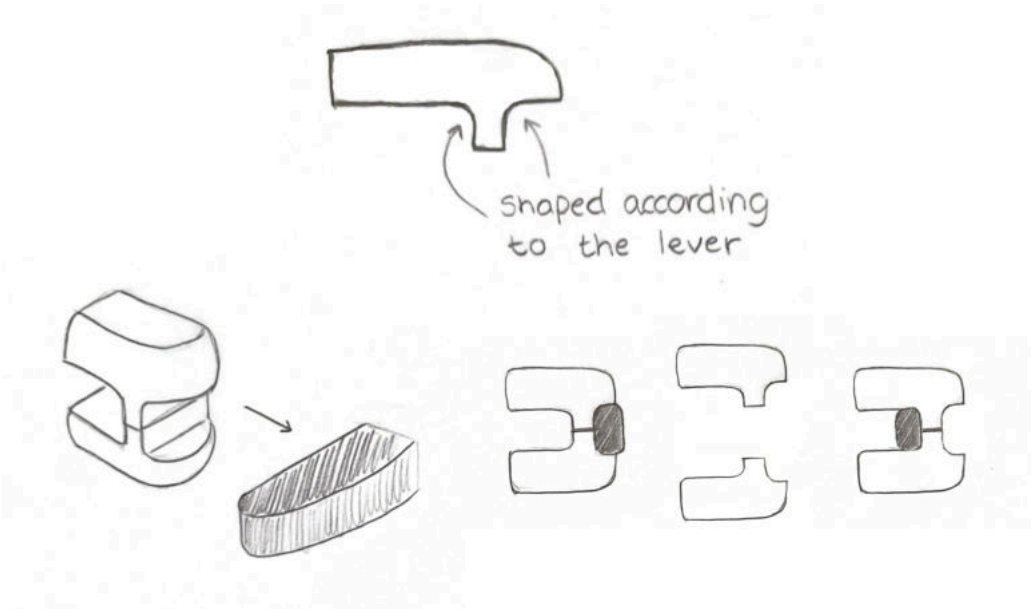


Figure 5.12: Sketches of the final concept

5.5 Final concept

The final concept of the Cobot fingers is presented in Figure 5.13 constructed to be compatible with the Onrobot finger gripper utilized by Elekta. This configuration allows for an effortless attachment, as the fingers smoothly slide onto the gripper and are securely fastened using the accompanying screws. The fingers are designed to operate on both sides of the CT box without the necessity for a finger switch. Consequently, this design optimizes operational efficiency and reduces the need for continual employee involvement.



Figure 5.13: Final concept

The fingers are tailored to the specific dimensions of the lever, to grip the lever securely when both rotating the lever and flexing the frame. The finger's manipulation of the CT box mimics human motion during testing, thereby ensuring results in accordance with those conducted through manual testing. The fingers grip the lever from opposing sides and flex the frame by grabbing the inside of the lever, mirroring the actions of human fingers, see figure 6.5.

To further enhance performance, the 3D-printed nylon fingers are coated with a soft plastic covering. This not only reduces friction but also protects the product against damage, while also simulating the softness of human fingers. See figure 5.14



Figure 5.14: Materials

5.6 Concept implementation

The Cobot's operation is controlled by code written in RoboticStudios. The primary emphasis during the code development has been on ensuring user-friendliness. Since the operators are handling and adjusting the code and Cobot during testing, it is important that they can perform these tasks without the need for additional expert aid, thereby enhancing overall efficiency and use of resources.

One cycle performed by the Cobot consists of three operations. Initially unlocking the lever, followed by the flex of the CT box frame 2 cm in an orthogonal direction to the frame side, and then locking the lever again. Each operation consists of several movements, designed to optimize the performance of the Cobot. When defining the movements of each operation, lead-through programming combined with jogging 3.1.6 the Cobot was used to achieve optimal positions. To complete the first operation of unlocking the lever the Cobot passes through three locations, see figure 5.15. The following operation of flexing the frame consists of four

gripper adjustments and three location changes. To finalize the cycle, the Cobot performs steps through the same locations as the initial operation, to reach its start location. The cycle, including the three operations, is illustrated in figure 5.15. After performing the desired number of cycles set as the target, the Cobot moves through 5 middle locations, transporting the Cobot arm to the opposite side to perform the same number of cycles there.

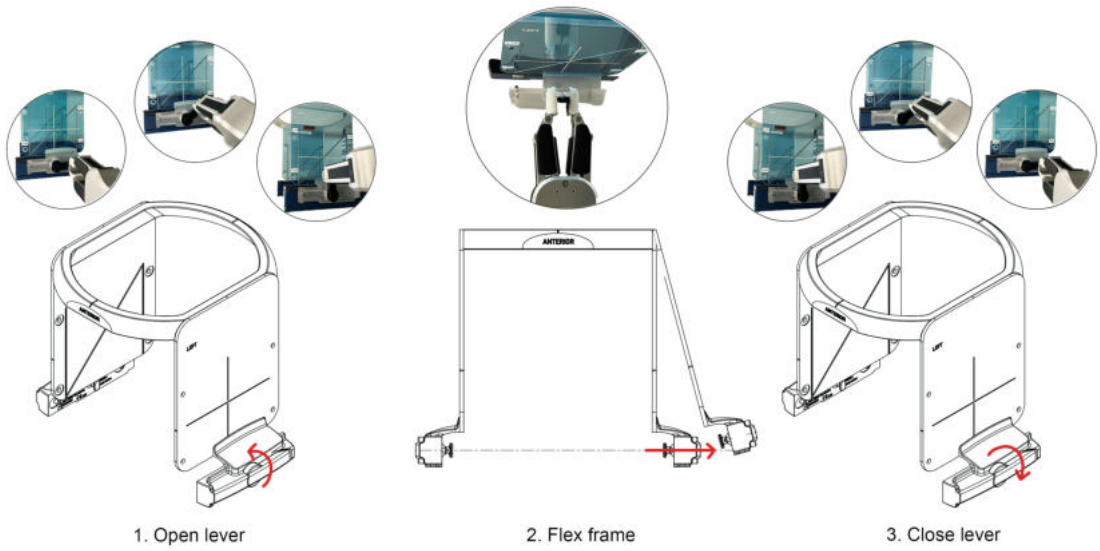


Figure 5.15: Cobot operations

5.6.1 RAPID programming challenge

After defining the cycle locations, a challenge arose as the Cobot halted at each designated position before proceeding with the next program instruction. This calls for an inefficient cycle process. To make the process more efficient the RAPID programming technical reference manual was scanned to find a suitable data type to streamline the process. Zone data was integrated into the program, it is used to specify how a position is to be terminated, i.e. how close to the programmed position the axes must be before moving toward the next position. This enables

a smoother reorientation and a cycle where the Cobot does not have to reduce velocity to carry out the reorientation. The defined locations, except start and finish positions, were therefore changed from stop points to fly-by points with a 20mm path zone in which the Cobot can proceed to the next position. [7] The whole RAPID program code is available in Appendix A.4

6 Result

The result presents the deliverables of this master's thesis on the implementation of Cobots for automating mechanical verification tests at Elekta. It emphasizes a strategic approach, encompassing test selection, gripper interface selection, finger customization, coding techniques, and a demonstration. The chapter combines ergonomic considerations, testing efficiency, and task complexity to guide optimal test automation decisions. It concludes by examining the benefits and challenges of Cobots, considering economic, environmental, and social implications. Lastly, this chapter provides insight into how Cobots revolutionize testing processes and contribute to Industry 4.0's goals of innovation and sustainability.

6.1 Cobot implementation guidelines

This section encompasses Cobot implementation guidelines, that guide Elekta through the steps of future Cobot implementations. These guidelines are developed based on insights gathered from the theoretical- -user- and market research combined with insights from the creative process of automating a mechanical verification process. The guidelines contain four areas of consideration:

1. Selection of mechanical verification test suitable for automation
2. Gripper selection
3. Finger development
4. Coding technique

6.1.1 Selection of mechanical verification test suitable for automation

Choosing a mechanical verification test suitable for automation involves a strategic approach guided by insights from user research, industry expertise, and ergonomic considerations. The key drivers for automating verification processes include eliminating ergonomic issues and enhancing testing efficiency. Based on the provided information, choosing a mechanical verification test suitable for automation can be facilitated through the following steps:

1. Identify Pain Points and Efficiency Gaps

The first step is to evaluate the test movements, especially cases of lifetime and wear, where repetitive and monotonous movements often occur. Begin by understanding the pain points in the test and inefficiencies in the current manual verification process. Identify tests where human involvement is particularly challenging due to repetitive actions and extended cycles. Cobots present a solution to these challenges by addressing ergonomic concerns and increasing testing efficiency.

2. Prioritize Safety Concerns

The next crucial step is to evaluate safety, particularly when working alone during high-risk hours or at night in the laboratory. Identify tests where safety hazards are a significant factor and could be addressed through automation, where the Cobot can replace human presence in such scenarios.

3. Evaluate Test Difficulty

Tests involving complex movements, challenging problem-solving, and significant waiting periods are prime candidates for automation. This aligns with the expertise of Cobots in precision and automation and can give more accurate results. Leveraging insights from industry experts and incorporating their recommendations can also further guide the selection process.

4. Analyze Limitations

Cobots have limitations, such as the inability to perform two-hand operations, which must be considered when selecting automation tests. The chosen verification test should align with Cobots' strengths in precision and automation, effectively addressing ergonomic challenges, safety risks, and efficiency gaps. With the help of the information in table 4.1, a comprehensive decision can be made.

Combine the three key pillars; ergonomic considerations, efficiency, and difficulty level, to make an informed decision of which verification tests are suitable for automation. Focus on tests that align with the strengths of Cobots, such as addressing ergonomic concerns and improving efficiency. This approach ensures that the selected test optimally benefits from automation, improving overall testing processes, employee well-being, and laboratory safety.

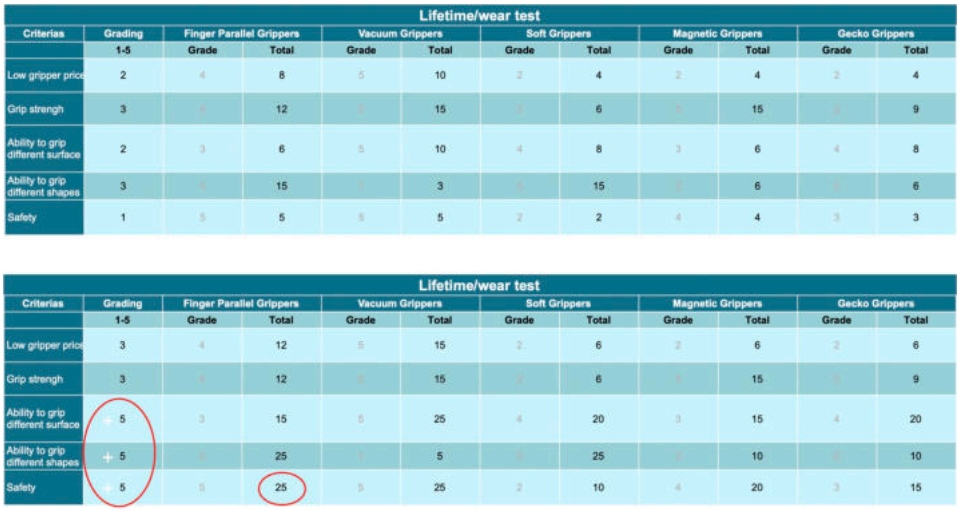
6.1.2 Gripper selection

The formulation of guidelines serves the purpose of simplifying the user experience with the Cobot, particularly given its new implementation at Elekta. Despite the Cobot's extensive automated capabilities, there exists significant variability in the testing procedures. Notably, the Elekta Cobot operates in conjunction with the testers, necessitating their active involvement in configuring the setup before testing. This entails creating the program in accordance with the specific test parameters, followed by the initiation of the program to initiate the test execution.

The initial step towards creating a program tailored to a particular test involves the selection of an appropriate gripper mechanism. To streamline this process, the creation of a matrix was undertaken, aimed at simplifying the selection of a gripper in alignment with the distinct demands of each test scenario. The underlying concept of this matrix revolves around the ability to commence from a standardized set of general test cases with diverse weighted criteria. Subsequently, using insights garnered from the gripper research, an evaluation of these cases has followed, predicated on the established criteria.

For the convenience of users, a document, provided in Appendix A.2, facilitates the selection of the most suitable gripper for the specific test and associated product. This document is also designed to accommodate potential modifications prompted by the product's specifications or changes in product test parameters. To illustrate, in cases involving exceptionally delicate products, the significance

of a gripper's capability to handle fragile items can be crucial, potentially giving different outcomes from the standard test cases, see 6.1.



Lifetime/wear test

Criterias	Grading	Finger Parallel Grippers		Vacuum Grippers		Soft Grippers		Magnetic Grippers		Gecko Grippers	
	1-5	Grade	Total	Grade	Total	Grade	Total	Grade	Total	Grade	Total
Low gripper price	2	4	8	5	10	2	4	2	4	2	4
Grip strengh	3	4	12	5	15	2	6	5	15	3	9
Ability to grip different surface	2	3	6	5	10	4	8	3	6	4	8
Ability to grip different shapes	3	5	15	1	3	5	15	2	6	2	6
Safety	1	5	5	5	5	2	2	4	4	3	3

Lifetime/wear test

Criterias	Grading	Finger Parallel Grippers		Vacuum Grippers		Soft Grippers		Magnetic Grippers		Gecko Grippers	
	1-5	Grade	Total	Grade	Total	Grade	Total	Grade	Total	Grade	Total
Low gripper price	3	4	12	5	15	2	6	2	6	2	6
Grip strengh	3	4	12	5	15	2	6	5	15	3	9
Ability to grip different surface	5	3	15	5	25	4	20	3	15	4	20
Ability to grip different shapes	5	5	25	1	5	5	25	2	10	2	10
Safety	5	5	25	5	25	2	10	4	20	3	15

Figure 6.1: The customized finger positioning during the test process

By this approach, testers are empowered to determine the gripper variant best suited to their unique testing requirements, leveraging the insights gained from the comprehensive research conducted within this project.

6.1.3 Finger development

Customizing fingers for specific tests and products offers advantages such as enhanced operational efficiency, safety, and accurate results. The Finger development can be essential to make a solution without the need to change the gripper. The development process used in this project involves a comprehensive evaluation of finger design factors. Based on the research and development in this project, a range of important factors have emerged, highlighting the strategic considerations for creating effective and specialized testing tools.

1. Make it simple

First and foremost, it's wise not to overcomplicate things; often, simplicity holds the key to the best solutions. At Elekta, the Cobot will be tasked with testing

a range of diverse products, which frequently arrive within specific intervals. Hence, it should strive to be a quick solution to uphold the efficiency of Cobot utilization. As a result of this project, it does not need to be complicated to get the work done as efficiently and safely as possible. This recognition underscores the significance of commencing with the original CAD files, see figure 6.2 of the Onrobot's finger, whenever feasible. This approach guarantees consistent mounting, ensuring a secure and reliable grip.

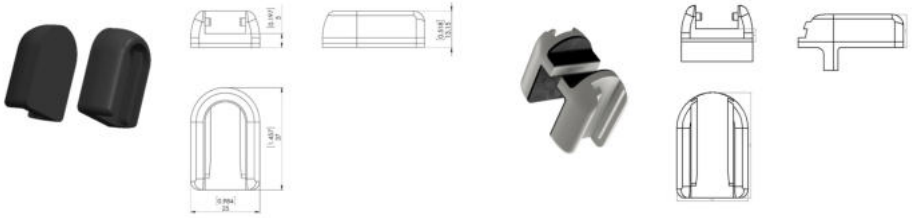


Figure 6.2: Original fingers and the customised vantage fingers

2. Should be easy to do it right

Ensuring user-friendly finger application and usage is crucial to prevent errors and product damage. This consideration gives significance due to the potential involvement of different testers initiating the tests. Simplified comprehension is facilitated by tailoring finger designs to match the product's specifications, like form, and incorporating suitable protective materials, see figure 6.3. Additionally, securing a stable application of the product in the designated work area contributes to a safer process.

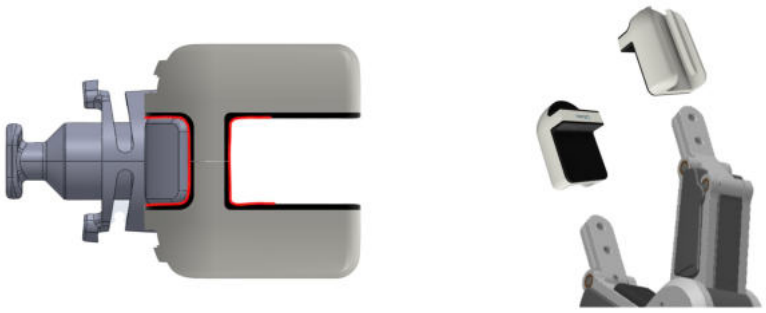


Figure 6.3: Tailored finger design

3. Minimize human interaction

The introduction of the Cobot for test automation presents an opportunity to enhance security by minimizing human intervention through carefully designed fingers. A key strategy to achieve this is by creating fingers that prevent the need for frequent switches during testing. Particularly, when tests encompass multiple gripping segments of a product, a universal finger design can prove beneficial, see an example of this in figure 6.4. This approach ensures the same set of fingers can adeptly address various product areas, streamlining the testing process.

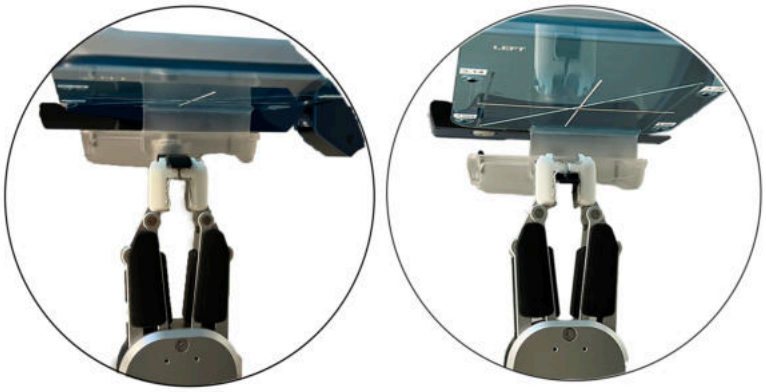


Figure 6.4: Creating universal fingers

4. Enhancing product integrity

As previously mentioned, incorporating a gentle material such as latex or soft plastic offers valuable protection against potential product damage or marks. This becomes especially significant for products destined for hospital use after testing, ensuring their new, factory appearance. Another convincing motive for material application is to optimize friction, which guarantees a secure grasp by the Cobot, minimizing the risk of grip loss and potential product harm caused by misalignment. Moreover, the inclusion of such material serves the purpose of emulating the tactile experience of human touch, enhancing the credibility of results by simulating real-world scenarios where individuals handle the product.

The development of Cobot gripper fingers involves a systematic approach encompassing attachment techniques and material selection for durability and friction. Additionally, it is important with ease of use, operational efficiency

through minimal component switches, gripping performance, and protection against product damage. The iterative design process helps refine the fingers, address challenges, and optimize performance. By prioritizing these factors, tailored fingers can facilitate seamless testing processes while safeguarding products and improving testing accuracy.

6.1.4 Coding technique

Code has been developed to conduct the test, this code enables smooth and effective movements. When creating a program it is best to combine a lead-through technique, jogging, and coding in RobotStudios. By creating the general movement with the Cobot it is easier to adjust the program to smooth movements and changing forces and speed. With the help of accompanying documents to Robotstudios, already made code is available that can be inserted and adjusted to the specific test and program.

6.2 Implementation

For this project Elekta had selected the gripper primarily for economic reasons, opting to utilize existing equipment. As mentioned earlier, a test was carefully chosen to maximize the capabilities of the finger gripper and Cobot, making it the preferred choice for both Elekta and their testers.

The code is developed to operate smoothly between defined locations, performing the lifetime test on the CT box. Ensuring the safety of the product undergoing testing is of great importance, and therefore, the Cobot program is designed to adjust its operation speed accordingly. During critical movements, the Cobot operates at a slower pace to minimize potential risks. In contrast, during transportation between critical operations, the Cobot operates at a faster speed to enhance overall efficiency and reduce unnecessary delays. This adaptive speed ensures a safe and efficient workflow while maintaining necessary levels of caution during critical phases.

The Cobot is programmed to execute the desired number of testing cycles by setting the target number accordingly. Upon completion of each cycle, the flex-pedant communicates the number of cycles completed on each side to the operator. Once the testing cycle reaches the predetermined target number, a message indicating ”the target has been reached” is conveyed to the operator. This process ensures intuitive, continuous, and efficient communication between the Cobot and the user.

Another aspect regarding product safety is the use of customized fingers to help ease the movement of gripping and flexing the frame safely. The fingers are customized so that they have a secure grip at all times. The cavity made in the fingers is designed with the exact curvature of the lever arm, see figure 6.5. This will guide the Cobot in the correct placement, and with the help of anti-slippage material, the finger will have a firm grip without slipping on the blank surface of the lever arm. To ensure a secure grip on the lever and prevent any risk of product loss due to high force and slippage, there is an additional cavity further back on the fingers with the same curvature as the lever arm. This allows for a secure flexing of the frame, see figure 6.5.

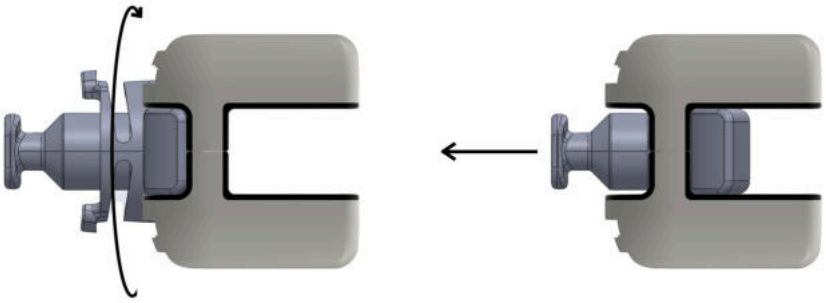


Figure 6.5: The customized finger positioning during the test process

The ultimate result is that the program and fingers created in this project have

been delivered to Elekta and used for real lifetime tests on the CT box and frame holder.

6.3 Validation

By implementing a Cobot and automating mechanical verification tests at Elekta, significant benefits are gained in the journey towards Industry 4.0. There are several advantages to the integration of automation, especially since it brings significant improvements in testing efficiency. Cobots can execute tests with a consistent speed and accuracy, reducing the time required for each verification cycle. This not only accelerates product development but also allows for more extensive testing. Cobots excel in precision and repeatability, ensuring that mechanical verification tests are conducted with minimal deviations. This consistency leads to reliable results and a higher degree of product quality, reducing the likelihood of defects or malfunctions.

Automation streamlines testing processes, reducing the need for human involvement in repetitive tasks. This, decreases the need for human resources and critical thinking, problem-solving, and decision-making, enhancing overall productivity. Automation also brings the elimination of ergonomic risks associated with repetitive and monotonous tasks. By relieving human workers from these tasks, Elekta can create a safer working environment and promote employee well-being. This enables round-the-clock operations with minimal human interaction. Automated verification processes utilizing Cobots can be scheduled to operate continuously, allowing for uninterrupted testing, even during the night.

Despite many advantages, there are disadvantages to consider. There is always an initial investment when integrating a Cobot, which also requires a significant investment in terms of technology, infrastructure, and training. The skills gap is a factor, the Cobot demands certain programming and robotic skills to conduct tests. Elekta has therefore the need to invest in training for testers to effectively

manage the Cobot.

The complex setups of product systems can also be a hindrance to efficiency since this requires time-consuming tasks such as integrating these systems seamlessly for accurate testing. Additionally connected to this is the limited adaptability and ability to perform certain tasks that require certain movements and decision-making. The automated system of a Cobot will also require maintenance and downtime for repairs and updates, ensuring the functionality of the systems. Lastly, even if the Cobot can execute tests with good accuracy this is not always the goal, especially when testing for a real-world environment. This can at some length be programmed but will not replace the human touch.

6.3.1 Sustainability

Social, economic, and environmental sustainability are three interconnected pillars that form the basis of a balanced and responsible approach to development, see figure 6.6.

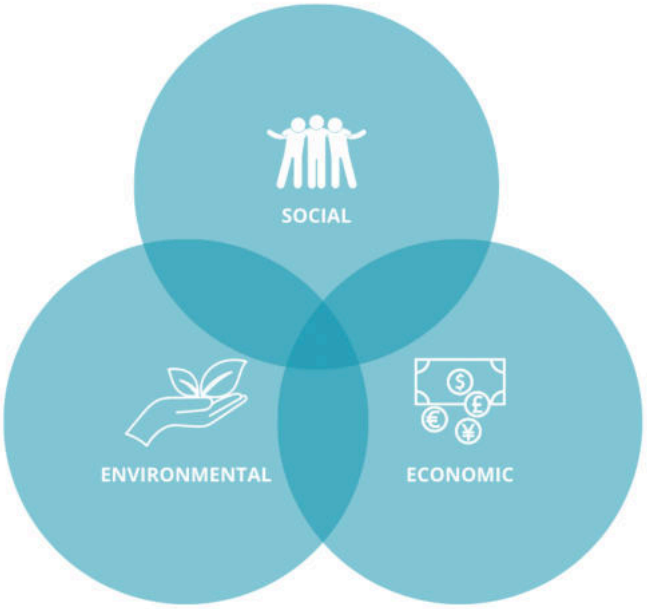


Figure 6.6: Illustrating the three interconnected pillars of social, economic, and environmental sustainability

The integration of Cobots at Elekta not only revolutionizes testing processes but also brings about significant sustainability implications across social, economic, and environmental dimensions. This integration towards Industry 4.0 and development at Elekta has several benefits that extend beyond the domain of efficiency and precision.

Social implications:

Two key aspects are central to the advancement of social sustainability: the implementation of the Cobot and the refinement of testing procedures. These two aspects primarily address the improvement of working conditions and safety measures at Elekta. The introduction of Cobots has a profound positive effect, notably by taking on monotonous, repetitive, and overall physically demanding tasks, thereby reducing ergonomic injuries. Accordingly, this shift contributes to developing a healthier and safety-oriented work atmosphere, ultimately fostering the overall well-being of the testers. However, it also raises an ethical aspect, that all automated tests will displace human testers, and have a potential loss of skilled testers.

Economic implications:

The economic implications of employing a Cobot are significant. The Cobot can function tirelessly, preventing the necessity for breaks and minimizing production downtime. Their capacity for continuous operation profoundly heightens efficiency across the testing procedures, thereby accelerating productivity and shortening the testing time. Furthermore, Cobots' ability to operate at nighttime extends the effective employment of tests.

Additionally, in Elekta's context, this proves advantageous as they face challenges in securing testers for all their required tests. The role of a tester is not a full-time job throughout the entire year, instead, it involves specific periods when tests come in. Given the sporadic availability of testers, this approach safeguards Elekta's dependency on the availability of testers, ultimately leading to enhanced efficiency during testing phases.

This enhanced resource utilization directly translates into elevated economic advantages, as testing can continue even in the absence of testers. Moreover, automation facilitates superior resource allocation, minimizing time and energy consumption. This, in turn, fosters cost savings and reduces the ecological impact linked with test processes.

Environmental implications:

From an environmental standpoint, Cobot technology offers several benefits. The main strength is Cobot's precision-driven automation which reduces the occurrence of errors, subsequently lowering the risk of defective items and minimizing waste. This efficiency not only restrains environmental impact but also diminishes the requirement for extra resources to rectify mistakes.

Moreover, Cobot's accurate and reliable testing procedures lead to products that meet strict quality standards. This, in turn, extends the lifespan of products and

decreases premature failures, fostering a more sustainable consumption pattern. The effect of these improvements contributes significantly to an overall reduction in energy consumption and the related emissions of greenhouse gases.

7 Discussion

Integrating a Cobot for automating the testing process at Elekta delivers a significant leap forward in testing efficiency and precision. This places Elekta at the forefront of technology within their industry. Enabling them to evolve consistently alongside technological advancements and utilize the benefits of Industry 4.0, mentioned in section 3.1.1.

However, this transition also raises an ethical consideration: the potential displacement of human workers by machines. While Cobots are designed for tasks requiring precision and repetition, their introduction into the testing process does pose ethical challenges. Challenges such as replacing human labor with robots. This raises consideration of Elekta's responsibility to facilitate fair transitions and provide essential retraining and support for affected employees. It also raises a broader ethical discussion on the societal impact, focusing on job security and economic stability. Achieving a balance between automation benefits and ethical concerns requires thoughtful reflection and strategic planning for a seamless transition into the future of work within Industry 4.0.

In the user, research carried out with industry professionals from Atlas Copco, as outlined in section 4.2, the significance of conducting a comprehensive pre-study was emphasized. Gaining an understanding of the background and ongoing needs is crucial when entering a new industry sector. This pre-study is intended to establish a solid groundwork for making well-informed decisions during the integration of robotics into a process that was previously manual. Over the long term, investing in this foundational research is expected to reduce risks, costs, time, and resource utilization. However, for these benefits to endure, it is important to consistently revisit and update the established foundation.

Another factor mentioned in the research of this project emphasizes the importance of using the ISO/TS 15066:2016 standard, in chapter 3.1.8 for ensuring the safe handling of Cobots. It is important to bear these standards in

mind during the development of new test processes. Doing so not only enhances the overall security of the testing procedures but also contributes significantly to the trustworthiness of both testing and Elekta's development. By aligning with these standards, Elekta ensures a framework that prioritizes safety and reliability in its industrial development and integration into the field of automation.

For future implementations, it is also important to consider the ergonomic risks associated with tasks that fall beyond the capabilities of the Cobot. For instance, objects with a weight of more than 5kg. As described in chapter 3.1.9, "the greater the force, the greater the risk". These objects set a higher risk factor for the operator thus they have a greater necessity for automation. In the context of expanding test automation, it is crucial to prioritize this concern to enhance the versatility of the Cobot and to create a safe and risk-free work environment in line with the ISO/TS 15066:2016 standard.

The incorporation of robotics in testing processes introduces concerns about the potential loss of essential human factors. While automation enhances efficiency, precision, and speed, it risks oversimplifying testing scenarios, neglecting human unforeseen variables and irregularities. The balance between automation and human involvement is crucial to preserving the reliability of testing outcomes in an increasingly automated landscape.

An intriguing question during the project was how the approach might have evolved with a longer duration. Several areas could have been approached differently to deliver more comprehensive and nuanced results. We could have refined the Cobot's fingers, exploring methods to mimic human touch and movement patterns. This includes introducing irregularities into the Cobot and research materials that closely replicate the human touch. The availability of more resources would have allowed us to conduct extensive experiments with various materials, enabling us to better replicate the nuances of human touch in our testing procedures. This would create a more authentic interaction with products, simulating real-life testing and usage scenarios. Additionally, dedicating more

time to understanding human movements when interacting with the products could have improved the replication of their actions in the Cobot test of the Vantage CT box. These enhancements, facilitated by an extended project timeline, would have elevated the quality of our results.

Finally, we must consider the factors that may have influenced and produced distinct outcomes within the project. Initially, Elekta had already selected the gripper before the project's start and desired its implementation before purchasing additional grippers. While we did explore alternative grippers to gain insight into their potential applications, the scope of this exploration remained somewhat theoretical. However, this part could have been more thoroughly developed, potentially yielding different results. Had we been able to conduct practical tests comparing the grippers across different test cases, we could have significantly enriched our findings. The results we achieved were shaped by all the elements discussed. The resources employed, time constraints, strategies, and knowledge obtained during the thesis.

7.1 Future work

In Elekta's upcoming initiatives, continuing our work would focus on enhancing the integration and sensitivity of the Cobot in upcoming test cases. This involves an exploration of gripper selection and testing, delving deeper into available grippers and their functionalities, and assessing specific test requirements. This approach allows Elekta to tailor their testing process to the unique characteristics of the products, yielding more precise outcomes and cost-effective solutions.

Furthermore, Elekta could continue to enhance data collection and refine the sensitivity of the Cobot's fingers. This involves an exploration of mimicking human behavior, studying human movement patterns, and exploring various materials to produce Cobot attachments. This pursuit contributes to irregular and sensitive operations delivering more accurate, efficient, and precise testing processes by incorporating real-world product usage scenarios.

Elekta's ongoing work should also involve developing additional customized Cobot fingers and creating a library of options based on specific testing requirements Additionally, we see a potential in designing a gripper or mechanism capable of executing two-handed movements, improving existing resources instead of requiring an entirely new Cobot.

As previously discussed with Elekta, we acknowledge the need to reassess the Cobot's power capacity, currently limited to handling 5 kg, even less when equipped with a gripper. This limitation presents an opportunity for Elekta to develop a more robust force lever arm, capable of exerting greater force. This innovation would expand product testing capabilities and reduce ergonomic risks associated with lifting and handling heavy objects.

Implementing the Cobot in Elekta's mechanical verification processes has demonstrated numerous opportunities and pathways for improvement. With this solid blueprint now in place, Elekta faces endless opportunities for advancement.

8 Conclusions

This master's thesis has successfully addressed Elekta's mechanical product verification challenges by implementing a Cobot into their product verification processes. Addressing issues of resource management, ergonomic and safety concerns, and inefficient procedures. Through comprehensive research, product development, and implementation guidelines, this thesis strategically implemented the Cobot, providing a systematic approach to product verification. This not only resolves existing challenges but also paves the way for upcoming automated operations.

Cobot implementation delivers numerous advantages, such as efficient resource allocation, reduced safety and ergonomic risks, and heightened testing precision. However, complex testing methods come with some considerations, including initial investment costs, ethical concerns about replacing human labor, and the ongoing need for human supervision. Despite these considerations, Elekta can use this thesis as a blueprint for Cobot implementation, optimizing verification procedures, and advancing into Industry 4.0.

In conclusion, integrating Cobots is of great importance to address ergonomic concerns, enhance efficiency, and stay competitive in the evolving industrial landscape. Our research showcases seamless Cobot integration into lifetime and wear test scenarios, indicating the potential for complementary tools to address more complex test systems. This holistic approach has the potential to automate Elekta's product verification process and position them as leaders in modern industrial technology.

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A

A Appendix

This section provides supplementary information to support and enhance the understanding of the main content.

A.1 Appendix - GANTT schedule

Presented below is the Gantt Schedule, which outlines the planned timeline that was followed for the successful completion of the tasks encompassed within this thesis.

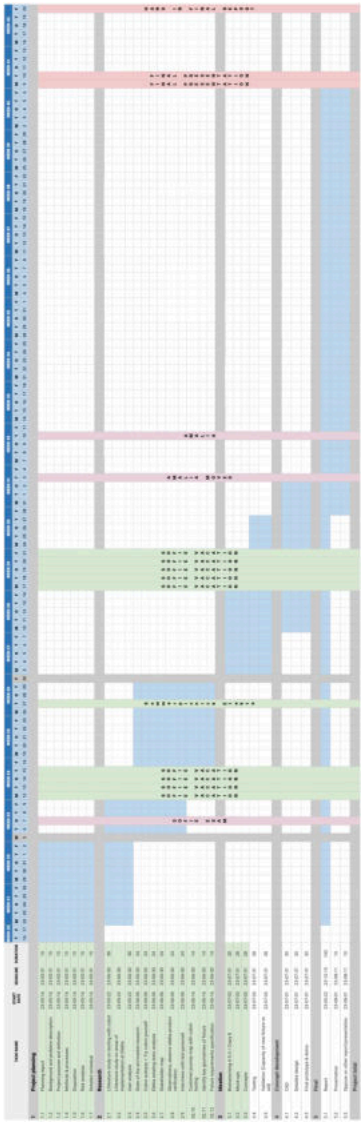


Figure A.1: GANTT schedule

A.2 Appendix - Interview guide

The interview guide presents the predetermined questions that served as the foundation during the interview sessions.

INTERVIEW GUIDE

Name:

Age:

Occupation:

Employment/qualifications

What is/was your employment at Elekta?

Hur länge har ni jobbat på Elekta?

Were there any qualifications needed for the job at Elekta?

Are there any safety courses that needed to be conducted before the testing/verification process?

Test

What types of tests and verification activities are commonly performed for Elekta's products? (e.g., functional testing, performance testing, safety testing)

What tests have you done at Elekta?

What are the key objectives of the testing and verification phase in the product development lifecycle at Elekta?

Are there specific methodologies or frameworks that you follow?

Do you have any freedom in how the testing process can be conducted?

How long does a test take?

What tools and technologies are utilized during the testing and verification process?

How do you ensure the availability and effectiveness of these tools?

How do you ensure that the testing process aligns with regulatory requirements and industry standards in the medical technology field?

Can you describe the typical testing and verification process at Elekta, from assignment to finished test?

Are there different degrees of difficulty in the tests?

What is the most challenging thing/procedure when testing?

Have you ever felt that a test is too challenging to complete?

How do you handle test data management and documentation to ensure traceability and reproducibility of test results?

How do you track and address defects or issues identified during testing?

Safety/ergonomics

Describe how you work with safety and ergonomics during a testing procedure.

Are there any specific training/ guidelines or awareness programs to educate testers about ergonomic risks?

What safety protocols or standards are followed during testing to avoid potential risks or hazards?

How do you ensure the safety of the testing environment?

Can you describe a situation where you encountered a safety concern during testing and how you addressed it?

Have you gotten any injuries/ discomfort from testing? mark on map.

What measures are taken to ensure that the products being tested meet safety regulations and compliance requirements?

Can you share any instances where ergonomic improvements in the testing process have led to increased productivity or reduced fatigue and discomfort for testers?

Automation

How does Elekta approach continuous improvement in the testing and verification process?

Can you share any experiences or strategies related to test automation at Elekta?

Are there any tests suitable for automation?

H

A.3 Appendix - Elimination gripper matrix

Presented below are three distinct elimination matrices for the lifetime/wear test, precision test, and displacement test.

Lifetime/wear test											
Criterias	Grading	Finger Parallel Grippers		Vacuum Grippers		Soft Grippers		Magnetic Grippers		Gecko Grippers	
	1-5	Grade	Total	Grade	Total	Grade	Total	Grade	Total	Grade	Total
Low gripper price	3	4	12	5	15	2	6	2	6	2	6
Grip strengh	4	4	16	5	20	2	8	5	20	3	12
Ability to grip different surface	4	3	12	5	20	4	16	3	12	4	16
Ability to grip different shapes	4	5	20	1	4	2	20	2	8	2	8
Safety (**droprisk**, sensors, camera)	5	5	25	5	25	2	10	4	20	3	15
Low gripper weight	5	3	15	3	15	2	10	3	15	4	20
Low product imprint and damage risk	5	4	20	3	15	5	25	5	25	5	25
Precision	5	5	25	2	10	4	20	4	20	4	20
Costumization	4	3	20	1	4	5	20	2	8	2	8
Object size	3	2	6	3	15	2	6	3	15	3	9
Object weight	4	3	12	5	20	1	4	4	16	2	8
Ability to handle dirt	3	5	15	1	3	5	15	1	3	4	12
Low operation costs	4	5	20	2	8	5	20	5	20	5	20
Total:			218		174		180		188		179

Figure A.2: Elimination gripper matrix for lifetime/wear test

Precision test											
Criteria	Grade	Finger Parallel Grippers		Vacuum Grippers		Soft Grippers		Magnetic Grippers		Gecko Grippers	
	1-5	Grade	Total	Grade	Total	Grade	Total	Grade	Total	Grade	Total
Gripper price	2	4	8	5	10	2	4	2	4	2	4
Grip strengh	4	4	16	5	20	2	8	5	20	3	12
Ability to grip different surface	4	3	12	5	20	4	16	3	12	4	16
Ability to grip different shapes	4	5	20	1	4	2	20	2	8	2	8
Safety (droprisk, sensors, camera)	5	5	25	5	25	2	10	4	20	3	15
Gripper weight	4	3	12	3		2	8	3	12	4	16
Product imprint and damage risk	5	4	20	3	15	5	25	5	25	5	25
Precision	5	5	25	2	10	4	20	4	20	4	20
Costumization	5	5	25	1	5	5	25	2	10	2	10
Object size	4	2	8	5	20	2	8	5	20	3	12
Object weight	5	3	15	5	25	1	5	4	20	2	10
Ability to handle dirt	3	5	15	1	3	5	15	1	3	4	12
Operation costs	4	5	20	2	8	5	20	5	20	5	20
Total:			221		177		184		194		180

Figure A.3: Elimination gripper matrix for precision test

Displacement test											
Criteria	Grade	Finger Parallel Grippers		Vacuum Grippers		Soft Grippers		Magnetic Grippers		Gecko Grippers	
	1-5	Grade	Total	Grade	Total	Grade	Total	Grade	Total	Grade	Total
Gripper price	2	4	8	5	10	2	4	2	4	2	4
Operation costs	4	4	16	5	20	2	8	5	20	3	12
Grip strengh	4	3	12	5	20	4	16	3	12	4	16
Ability to grip different surface	3	5	15	1	3	5	15	2	6	2	6
Ability to grip different shapes	3	5	15	5	15	2	6	4	12	3	9
Safety (droprisk, sensors, camera)	5	3	15	3	15	2	10	3	15	4	20
Gripper weight	4	4	16	3	12	5	20	5	20	5	20
Product imprint and damage risk	5	5	25	2	10	4	20	4	20	4	20
Precision	5	5	25	1	5	5	25	2	10	2	10
Costumization	4	2	8	5	20	2	8	5	20	3	12
Object size	4	3	12	5	20	1	4	4	16	2	8
Object weight	4	5	20	1	4	5	20	1	4	4	16
Ability to handle dirt	3	5	15	2	6	5	15	5	15	5	15
Total:			202		160		171		174		168

Figure A.4: Elimination gripper matrix for displacement test

L

M

A.4 Appendix - Cobot code

Presented below is the Code programmed to operate the Cobot during the lifetime test of the Lexell Vantage CT Fiducial box and frame holder.

Kommented code Lexell Vantage CT Box

```
MODULE Wizard

Positions Left side
PERS robtarget StartL:=[[466.979614257813, 392.834686279297, 118.23900604248],
[0.00571453012526035, 0.680623829364777, -0.010404103435576, -0.732536852359772],
[0, 0, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

PERS robtarget Location1L:=[[466.707733154297, 431.981079101563,
120.003051757813], [0.230828791856766, -0.626963794231415, -0.240933284163475,
0.703978419303894], [0, 0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];

PERS robtarget Location2L:=[[468.312103271484, 450.461212158203,
155.530410766602], [0.374732494354248, -0.567410230636597, -0.407928645610809,
0.609274506568909], [0, 0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];
VAR num Target:=2;
VAR num CounterRight:=0;

PERS robtarget Location3L:=[[469.055511474609, 452.789306640625, 181.383193969727],
[0.488303303718567, -0.47207099199295, -0.532896161079407, 0.504708409309387], [0,
0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];
PERS robtarget OutR:=[[-361.885040283203, 451.291625976563, 164.519897460938],
[0.527503192424774, 0.503164708614349, 0.475409388542175, 0.492495208978653], [1,
-1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

PERS robtarget OutL:=[[469.894256591797, 452.880523681641, 181.062316894531],
[0.488296985626221, -0.471958667039871, -0.532817959785461, 0.504902064800262],
[0, 0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

PERS robtarget InL:=[[451.42529296875, 452.772888183594, 178.060073852539],
[0.488282889127731, -0.472024321556091, -0.532804548740387, 0.504868447780609],
[0, 0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

Positions Right Side
PERS robtarget StartR:=[[-360.355346679688, 404.327728271484, 123.083793640137],
[0.725459456443787, 0.0147866616025567, 0.688106000423431,
0.00000554587677470408], [1, -1, -3, 0], [9000000000, 9000000000, 9000000000,
9000000000, 9000000000, 9000000000]];
```

```
PERS robtarget Location1R:=[[-360.682434082031, 433.639801025391,
139.23600769043], [0.689438760280609, 0.215668648481369, 0.665886521339417,
0.186430752277374], [1, -1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];

PERS robtarget Location2R:=[[-360.810607910156, 447.485900878906,
151.218078613281], [0.615320980548859, 0.392302840948105, 0.573637306690216,
0.372046709060669], [1, -1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];

PERS robtarget Location3R:=[[-361.858581542969, 451.376373291016,
163.249938964844], [0.527407169342041, 0.503282070159912, 0.475375652313232,
0.492510765790939], [1, -1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];

PERS robtarget Out1R:=[[-374.844421386719, 451.382019042969, 163.231750488281],
[0.527424097061157, 0.503273785114288, 0.475370794534683, 0.492505759000778], [1,
-1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

PERS robtarget InR:=[[-341.883361816406, 451.298767089844, 164.539657592773],
[0.527494668960571, 0.503182828426361, 0.475406408309937, 0.492488771677017], [1,
-1, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];

PERS robtarget Out1:=[[476.669036865234, 452.791900634766, 178.019439697266],
[0.488322466611862, -0.472069054841995, -0.532869637012482, 0.504719614982605],
[0, 0, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];
PERS robtarget Grip:=[[467.789245605469, 404.564392089844, 209.452911376953],
[0.00453514698892832, 0.707939267158508, 0.0200461875647306,
-0.705974221229553], [0, 0, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];
PERS robtarget Middle0:=[[511.387359619141, 392.799652099609, 118.150779724121],
[0.00571438483893871, 0.680580258369446, -0.0104445610195398,
-0.732576787471771], [0, 0, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];
```

Positions between left and right side

```
PERS robtarget Middle1:=[[571.260192871094, 442.007202148438, 601.599365234375],
[0.00231709401123226, -0.99985283613205, 0.00982260517776012,
0.0138737224042416], [0, 0, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000,
9000000000, 9000000000]];
PERS robtarget Middle2:=[[28.5783195495605, 493.531341552734, 714.923706054688],
[0.0161227975040674, 0.705051779747009, 0.708972334861755,
0.000499043497256935], [0, -1, -3, 0], [9000000000, 9000000000, 9000000000,
9000000000, 9000000000, 9000000000]];
```

```
PERS robtarget Middle3:=[[-262.559753417969, 466.192321777344, 696.520263671875],
[0.00270634773187339, 0.0162267051637173, 0.999820351600647,
-0.00941560324281454], [1, 0, -3, 0], [9000000000, 9000000000, 9000000000,
9000000000, 9000000000, 9000000000]];
PERS robtarget Middle4:=[[-403.005126953125, 404.770294189453, 125.655860900879],
[0.722856402397156, 0.0146278813481331, 0.69083708524704, 0.00296328775584698],
[1, -1, -3, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];
PERS robtarget Turn:=[[527.202575683594, 392.026428222656, 217.773986816406],
[0.0184686649590731, -0.703239798545837, -0.0120482984930277, 0.710610687732697],
[0, 0, -2, 0], [9000000000, 9000000000, 9000000000, 9000000000, 9000000000,
9000000000]];
VAR num CounterLeft:=0;
```

Main Program

```
PROC main()
   MoveJ Middle0, v50, fine, tool0; %Setting the start position
IF CounterLeft = 0 THEN %Setting the gripper width to the right size, at the first cycle.
   OR_RGX_MOVE 6,50,1;
 ENDIF
%Starting the cycle on the Left side of the CT Box
WHILE CounterLeft <= Target DO
   MoveJ StartL, v50, fine, tool0;
   MoveJ Location1L, v200, z20, tool0;
   MoveJ Location2L, v200, z20, tool0;
   MoveJ Location3L, v200, fine, tool0;
   OR_RGX_MOVE 30,100,1;
   MoveJ InL, v200, fine, tool0;
   OR_RGX_MOVE 7,100,1;
   MoveJ OutL, v50, fine, tool0;
   MoveJ InL, v50, fine, tool0;
   OR_RGX_MOVE 30,100,1;
   MoveJ Location3L, v200, fine, tool0;
   OR_RGX_MOVE 6,75,1;
   MoveJ Location2L, v200, z20, tool0;
   MoveJ Location1L, v200, z20, tool0;
   MoveJ StartL, v200, fine, tool0;
   CounterLeft := CounterLeft+1; %Adding one cycle to the counter when the cycle is done.
   TPWrite "Number of cycles left: "\Num:=CounterLeft;  %Printing number of cycles
   IF CounterLeft = Target THEN
      TPWrite "Target has been reached on the left side!"; %Message when the target is
reached
%Moving fron the Left to the right side
      MoveJ Middle0, v200, fine, tool0;
      MoveJ Middle1, v500, z20, tool0;
      MoveJ Middle2, v500, z20, tool0;
      MoveJ Middle3, v500, z20, tool0;
      MoveJ Middle4, v500, fine, tool0;
```

```
%Starting the cycle on the Right side of the CT Box
      WHILE CounterRight <= Target DO
         MoveJ StartR, v50, fine, tool0;
         MoveJ Location1R, v200, z20, tool0;
         MoveJ Location2R, v200, z20, tool0;
         MoveJ Location3R, v200, fine, tool0;
         OR_RGX_MOVE 30,100,1;
         MoveJ InR, v200, fine, tool0;
         OR_RGX_MOVE 7,100,1;
         MoveJ OutR, v50, fine, tool0;
         MoveJ InR, v50, fine, tool0;
         OR_RGX_MOVE 30,100,1;
         MoveJ Location3R, v200, fine, tool0;
         OR_RGX_MOVE 6,75,1;
         MoveJ Location2R, v200, z20, tool0;
         MoveJ Location1R, v200, z20, tool0;
         MoveJ StartR, v200, fine, tool0;
         CounterRight := CounterRight+1; %Adding one cycle to the counter when the cycle is
done.
         TPWrite "Number of cycles right: "\Num:=CounterRight;
         IF CounterRight = Target THEN
            TPWrite "Target has been reached on the right side!";  %Message when the target
is reached
%Moving from the Right side to the left
            MoveJ Middle4, v500, fine, tool0;
            MoveJ Middle3, v500, z20, tool0;
            MoveJ Middle2, v500, z20, tool0;
            StopMove;
         ENDIF

      ENDWHILE
   ENDIF
ENDWHILE
ENDPROC
ENDMODULE
```

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